

RESEARCH ARTICLE

Optimizing Stroke Recognition With MediaPipe and Machine Learning: An Explainable AI Approach for Facial Landmark Analysis

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This work was supported by the Researchers Supporting Project, King Saud University, Saudi Arabia, under Project RSPD2025R951.

ABSTRACT Early detection of stroke is critical for improving survival rates and recovery. This study presents an innovative approach to stroke diagnosis through the analysis of facial landmarks combining MediaPipe's facial landmark detection with advanced machine learning models' Random Forest (RF), Extreme Gradient Boosting (XGB), and Categorical Boosting (CB). A comprehensive dataset of stroke and non-stroke facial images are curated, with MediaPipe extracting 228 facial landmarks from key regions affected by stroke-related asymmetry. Explainable AI (XAI) techniques are applied to enhance model interpretability, allowing for a deeper understanding of the most significant facial regions contributing to stroke prediction. The machine learning models are individually optimized through hyperparameter tuning, and further we propose a Multimodal Voting Classifier (MVC). By aggregating the predictions through majority voting or weighted averaging, the system reduces the likelihood of incorrect classifications that might arise from a single model's limitations. This results in a more stable and generalized model with improved diagnostic accuracy, as demonstrated by its 94.75% accuracy, than any individual model's performance. Feature importance analysis, facilitated by XAI, identified key facial regions—such as the eyes, cheeks, and lips—as critical indicators of stroke, significantly improving the model's diagnostic precision. Additionally, t-SNE visualization is employed to provide insight into classifying the stroke and non-stroke cases, reinforcing the model's robustness. Real-time metrics show the model's efficiency and adaptability across diverse hardware, enabling practical clinical integration. The proposed system offers a cost-effective, non-invasive diagnostic tool that can be implemented in real-time, potentially improving accessibility to stroke diagnosis in remote and resource-limited areas.

INDEX TERMS Explainable AI, facial landmarks analysis, feature importance, machine learning, MediaPipe, stroke recognition.

I. INTRODUCTION

Stroke is a universal health problem, and its impact is high on the scale of morbidity, mortality, and economic burden. The efficiency of stroke recovery is highly dependent on the prompt and accurate identification of stroke indicators.

The associate editor coordinating the review of this manuscript and approving it for publication was Filbert Juwono¹.

However, current technical methods of early diagnosis from facial images require specialized equipment and expert interpretation. Previous studies have applied machine learning techniques to this problem with varying degrees of success, but challenges remain in optimizing both the accuracy and interpretability of these models.

Stroke death rate elevated from 38.8 per 100,000 people in 2020 to 41.1 per 100,000 in 2021, according to a survey by

the Center for Disease Control and Prevention [1]. Ischemic stroke affects 15 million people worldwide and financial and mental costs are high on families and communities, as stated in a report by the World Health Organization (WHO) of Eastern Mediterranean region [2]. Moreover, Close to 34% of global healthcare expenditure is spent on stroke [3]. In spite of the advancements in the medical field, early recognition of ischemic stroke remains challenging, particularly in remote or under-resourced areas.

Stroke detection systems currently use Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) scans but both can be expensive, time consuming and they might not be available in certain regions [4]. This issue is addressed in the proposed system which does so by analyzing facial images for asymmetries and other stroke related facial attributes in real time with the help of machine learning algorithms for facial landmark detection. It combines MediaPipe [5], which is a reliable facial landmark detection system, to extract facial landmarks. This approach not only decreases the cost of imaging devices but also enables quick, on-the-spot screening.

Three sophisticated machine learning classification algorithms: Random Forest(RF), Extreme Gradient Boosting(XGB), and Catagorical Boosting(CB). The effectiveness of these models in detecting stroke symptoms based on facial features is thoroughly assessed.

This paper introduces a novel approach to diagnosing facial palsy through the application of machine learning and computer vision techniques. Our primary contributions are as follows:

- **Comprehensive Dataset Collection and Augmentation:** A diverse dataset of 1000 facial images (500 with stroke symptoms, 500 without) is curated from publicly available sources, ensuring a balance between positive and negative cases. The models are made robust by applying data augmentation techniques: rotation, flipping, and scaling. The dataset is expanded in size to strengthen the model's capability to pick out subtle stroke signs for other disparate demographic groups.
- **Advanced Facial Landmark Extraction:** Using MediaPipe Face Mesh, 228 key facial landmarks are selected out of the available 468 by focusing on regions most indicative of stroke or facial palsy-related anomalies. This refined method aims to minimize computational overhead while maximizing diagnostic relevance, by isolating those face areas indicated for stroke: asymmetry or drooping [6].
- **Intense Model Optimization Techniques:** The baseline models are thoroughly optimized using Ray Tune, Randomized Search, and Grid search in order to find the best performing hyperparameters. In particular, these techniques are use iteratively and are additionally fine tuned through manual calibration, post optimization. This leads to more performance and stability.
- **Multimodal Voting Classifier:** Through combining the strengths of RF, XGB, and CB models, a robust

Multimodal Voting Classifier (MVC) is created. This ensemble technique improves predictive performance by utilizing the best features of each model, which lead to increased accuracy and reliability in stroke detection.

- **Performance Evaluation and Validation:** The model goes through intense testing across various datasets. Its effectiveness is validated by key performance indicators such as accuracy sensitivity and specificity.
- **Explainable AI (XAI) Integration:** By emphasizing the importance of particular face regions in stroke detection, XAI approaches offer transparency and enable medical practitioners to validate the model's conclusions. Feature importance techniques for RF, XGB, and CB further refine the model by identifying the most effective facial landmarks.
- **Dimensionality Reduction using t-SNE:** t-Distributed Stochastic Neighbor Embedding (t-SNE) is employed for dimensionality reduction and visualization of high-dimensional data. This technique facilitates the interpretation of complex datasets and enhances the clarity of the classification process.
- **Real-time Performance of the model:** Real-time performance evaluation is important for evaluating the robustness of machine learning models in clinical contexts. In the study, Python based Graphical User Interface(GUI) is used to deploy the proposed MVC in a python application to process user input images to diagnose stroke while measuring different performance metrics such as the preprocessing time, inference time, and the total time taken. By testing on a range of hardware systems, the analysis highlights the model's adaptability, efficiency, and feasibility for the implementation into real-world workflows, ensuring reliability in time-sensitive scenarios.

The rest of the paper is organized as follows: Section II presents the literature review, Section III details the methodology, Section IV includes the experimental results analysis, section V describes the discussion of the study and finally, section VI concludes the results.

II. LITERATURE REVIEW

In recent years, there has been significant interest in leveraging image analysis techniques and machine learning algorithms for medical diagnosis, particularly in identifying stroke symptoms and facial paralysis. Various studies have explored these methods, each revealing notable limitations that impact their practical application.

Several studies have demonstrated the potential of machine learning in detecting stroke symptoms and facial paralysis using video and image analysis. One study utilized a combination of Histogram of Oriented Gradients (HOG) and Adaptive Boosting to diagnose facial palsy, but faced challenges such as lower accuracy in detecting leg paresis and ataxia, exclusion of certain algorithms due to long training times, and potential overfitting due to high accuracy in

controlled environments [7]. A study focused on automatic gesture stroke detection using supervised learning algorithms to identify facial and hand features, revealing limitations such as dependency on skin color detection, lack of ground truth data, and challenges with varying lighting and environmental conditions [8].

MediaPipe and machine learning are employed to detect stroke symptoms. The study highlighted issues such as the need for external validation, small sample size, and potential bias from manual video selection and labeling [9]. Similarly, research using OpenCV and Dlib libraries to develop a facial expression detection model for stroke patients showed promise but was limited by a small sample size, scope limitations focusing only on facial expression features, and the need for manual monitoring in real-time applications [10]. Utilizing the MediaPipe framework, specifically the Blaze-Face and FaceMesh models, for detecting facial landmarks in images containing facial paralysis. The main challenge faced is ensuring the generalizability and robustness of the system across a wide range of facial paralysis cases with varying degrees of severity and facial expressions [11].

Another research, “auto-eFACE” an automated facial palsy assessment tool enhanced by machine learning, utilized the Emotrics software for facial landmark localization and auto-eFACE for scoring facial metrics [12]. However, the study faced significant limitations such as technical difficulties in consistent landmark detection, generalizability issues due to a small sample size, incomplete facial assessment, the necessity of manual clinician intervention, and questions about the clinical significance of minor asymmetries detected. Computer vision techniques have also been explored for health assessment. One study used MediaPipe and penalized Linear Discriminant Analysis (pLDA) to sort films into groups based on weak facial features. Despite showing potential, the study faced limitations such as the need for further validation in real-world settings, small sample size and potential bias introduced by manual video selection. Another study made use of computer vision library called OpenCV and Dlib to create a cascade of regression trees technique that was trained on the 300-W dataset. It revealed problems related to limited expression variation, demographic bias on the used training dataset, and the requirement for validation in various clinical contexts [13].

A major field of interest in the evaluation of facial paralysis has been facial landmark detection. In one study, Support Vector Machines (SVMs) were implemented for classification and Active Shape Models (ASMs) were used to extract facial landmarks but the work encountered difficulties in spite of showing high accuracy, including scarce dataset availability, worries about overfitting, and the need for more extensive classifier validation and comparison in clinical situations [14]. CB is used to detect facial paralysis [15]. Data preprocessing is a step in the methodology that includes cleaning data to remove missing values and inconsistencies, correcting class imbalances, choosing pertinent features,

and dividing data into sets for training and testing. CB is used to oversee categorical data by transforming features to numerical representations while training. The study used criteria including precision and Area Under the Curve (AUC) to compare CB with other algorithms such as RF. However, the study has several drawbacks, including issues with generalizability brought on by reliance on a particular dataset, lack of external validation, possible bias in choosing features, model complexity that impedes clinician understanding, and an absence of discussion regarding clinical implementation.

A research applies voting classifier which is an ensemble learning technique used for stroke prediction [16]. Through Hard Voting (the majority class selection) and Soft Voting (an average of predicted probability), it integrates predictions from several models to improve accuracy. 20% was set aside for testing and the remaining 80% for training. Algorithms such as RF, Decision Tree, and Logistic Regression are trained alongside the Voting Classifier. The robustness of stroke detection results is enhanced by the Voting Classifier by combining predictions from these models. The models’ application in real-time clinical situations is limited by the use of a fixed dataset, and their dependence on just one data set raises questions regarding generalizability. These are some of the main constraints. Furthermore, the absence of feature selection and potential overfitting—particularly with the RF algorithm—highlight the necessity for additional study to strengthen the resilience and dependability of such models in a variety of real-world circumstances.

Another method for diagnosing facial palsy that have relied on Convolutional Neural Networks (CNN) had problems such poor face posture, trouble distinguishing real cases apart from mimicry, and ethical issues with data collecting [17]. SVM with an RBF kernel and K-Nearest Neighbours (KNN) for classification is employed by many researches for extracting regional information from photographs in order to diagnose facial palsy. While showing potential, limitations of the study included perhaps a small sample and need for use more fully selected features and validation in real clinical settings [18].

A system for identifying facial paralysis using machine learning and real-time facial animation units extracted from the Kinect V2 sensor have employed hand-crafted features and deep learning-based approaches for 5-categorical severity classification. This method encounters limitations like challenges with accessories and inter-personal variations, drawbacks of 2D imaging techniques, high costs, and discomfort related with 3D motion capture systems, and sensitivity to noisy data and overfitting in classification approaches. Systems using Kinect 1.0 face issues with limited FAUs, instability of FAU values in real-time, and inaccurate tracking of facial features [19]. The development of the facial paralysis grading system involved the utilization of the Kinect v2 sensor with SDK 2.0 for data acquisition and analysis. It focused on the automatic detection of facial features, quantitative static and dynamic symmetrical analysis of the face, and grading the performance of achieving facial

functions for both sides individually. Key limitations include ensuring the system's generalizability across diverse facial paralysis cases and the need for further validation to confirm the robustness and clinical applicability of the approach [20].

These studies collectively underscore the potential of computer vision and machine learning in medical diagnostics while highlighting areas for further research and improvement to enhance their clinical applicability and reliability. Table 1 shows a summary of the state-of-the-art research.

III. METHODOLOGY

The study progresses through three stages of model training: baseline, enhanced, and XAI-refined enhanced models with its primary dataset. It undergoes data preprocessing at different levels. In the baseline training RF, XGB, and CB are trained without hyperparameter tuning, establishing a reference for performance. The proposed model combines these three models as MVC. The enhanced models undergo hyperparameter tuning using Ray Tune, improving the models' robustness and generalizability. In the final stage, XAI-enhanced models are trained, where XAI techniques are applied, refining the models by focusing on the most important facial features for stroke detection, resulting in improved interpretability and reliability in predictions. The details of the dataset preprocessing and the three-stage model training are presented in the following sections. Fig. 1 shows the overall process of extracting the facial landmarks and model training.

A. DATA COLLECTION AND PREPROCESSING

In the study, a primary dataset for the proposed system is created. A diverse collection of facial images is curated from various sources including public repositories such as Kaggle and Google Images [21]. The dataset consists of a total of 1000 RGB images, among which 500 images are of individuals diagnosed with stroke, and 500 images of individuals without stroke symptoms [22]. Diversity in terms of age, gender, and ethnicity is ensured to create a representative sample. Fig. 3 shows the overall process of dataset collection.

Data augmentation techniques such as scaling, flipping, and resizing are applied to enhance the dataset's diversity and the model's generalization capabilities.

In order to ensure diversity in the dataset, a pre-trained model: DeepFace is utilized which is a state-of-the-art facial recognition and analysis framework that uses deep learning techniques for multiple facial attribute predictions [23]. Ethnicity, gender, age can be precisely identified through analyzing facial landmark patterns.

The collected images are analyzed using DeepFace to determine the percentage distribution of ethnicities in order to make sure a representative sample across various groups. Individual age group categories found during analysis utilize the model to classify participants as young adult or middle-aged or senior citizens. The study assesses gender

distribution to ensure representative gender equality exists among the participants.

The results of these analyses are on Table 2 for clarity. This step not only validates the dataset's diversity but also ensures that the model trained on this dataset will perform reliably across a wide range of demographic variations. Figure 2 illustrates a sample collage of stroke-affected individuals from our dataset, highlighting the diversity in terms of ethnicity, age, and gender distribution. This visual representation further reinforces the dataset's inclusivity and its potential for generalization across different demographic groups.

Augmentation methods such as flipping images are utilized to increase the variability in the dataset while preserving the integrity of facial features, leading to 1000 real images and 1000 augmented images. The 'Transpose' function from the PILLOW library is utilized to flip images. Emphasis is placed on selecting images with optimal lighting conditions and focusing on elderly individuals, who are more prone to strokes, to ensure the dataset's relevance and accuracy.

1) DATA CLEANING

In this study, rigorous data cleaning procedures have been implemented to ensure the integrity and quality of the dataset. Images that are low-quality or not aligned properly within the frame are discarded. In addition, the images are resized and cropped to show only the face in a frame. This step is crucial to maintain consistency and reliability throughout the dataset, thus minimizing the risk of bias or inaccuracies during model training and evaluation.

228 2D key points in regions relevant for detecting stroke symptoms. (a) Lips (43 indexes), (b) Eyes (114 indexes), (c) Eyebrows (29 indexes), (d) Nose (4 indexes), and (e) Cheeks (38 indexes) as shown in Fig. 4.

2) FACIAL LANDMARK EXTRACTION

MediaPipe Face Mesh is used to extract specific facial landmarks [24]. At first, a total of 468 landmarks are extracted from facial images, and the 10 silhouette landmarks are excluded. With the remaining 458 landmark values, 228 2D keypoints are created in (y/x) format from the available landmarks. These features include 228 2D key points which accumulate in 29 regions relevant for detecting stroke symptoms, such as the lips (43), eyes (114), eyebrows (29), nose (4), and cheeks (38) as shown in Fig. 4. The lightweight model architectures of Blaze Face and GPU acceleration utilized by MediaPipe enable real-time performance, making it suitable for applications that require fast and responsive landmark detection [25]. The model configuration is shown in Fig. 5.

3) NORMALIZATION OF FACIAL LANDMARKS

Normalization of facial landmarks is crucial for standardizing the scale and position of features across different faces, which improves the accuracy and robustness of the model. The

TABLE 1. Summary of state-of-the-art research on facial paralysis diagnosis using machine learning techniques.

Reference	Contribution	Limitation
Jalo et al. [7]	Utilized HOG and AdaBoost in diagnosing facial palsy.	Recognition of leg paralysis poses challenges; complex tasks in recognizing stroke symptoms beyond facial paralysis.
Gebre et al. [8]	Employed supervised learning algorithms to identify facial and hand features distinguishing stroked frames from those without strokes.	Challenges in skin color detection and potential misidentification.
Aldridge et al. [9]	Utilized HoG traits and penalized Linear Discriminant Analysis (pLDA) to sort films into groups based on weak facial features.	Need for external validation in independent datasets and real-world scenarios to confirm the algorithm's generalizability; ethical concerns regarding the lack of informed consent from video participants.
Praditsangthong et al. [10]	Developed facial expression recognition using OpenCV and Dlib.	Potential limitations in handling diverse facial features and expressions.
Veeravalli et al. [11]	Utilized the MediaPipe framework for detecting facial landmarks in images containing facial paralysis.	Ensuring the generalizability and robustness of the system across a wide range of facial paralysis cases with varying degrees of severity and facial expressions.
Miller et al. [12]	Developed the "auto-eFACE," an automated facial palsy assessment tool enhanced by machine learning.	Technical difficulties in consistent landmark detection, generalizability issues due to a small sample size, incomplete facial assessment, the necessity of manual clinician intervention, and questions about the clinical significance of minor asymmetries detected.
Guarín et al. [13]	Utilized a cascade of regression trees algorithm for facial landmark detection, achieving high accuracy by fine-tuning with specific datasets.	Demographic and recording condition biases in the training dataset, which may lead to prediction errors and reduced model accuracy across diverse patient populations and clinical settings.
Vrochidou et al. [14]	Employed ASMs and SVMs for facial asymmetry detection.	Limited dataset availability, overfitting concerns, and imbalanced datasets.
Amsalam et al. [17]	Diagnosed facial palsy using CNNs to distinguish healthy from damaged faces.	Accuracy loss in poor face placement and generalization limitations.
Parra et al. [18]	Developed a system using SVM with RBF kernel and KNN for facial paralysis diagnosis.	Limited sample size and need for broader classifier comparisons.
Gaber et al. [19]	Utilized machine learning and FAUs from Kinect V2 sensor for facial paralysis detection, demonstrating high resilience and reliability.	Limitations in handling diverse facial features and expressions; questions about the clinical significance of minor asymmetries detected.
Gaber et al. [20]	Developed a facial paralysis grading system using the Kinect v2 sensor with SDK 2.0 for data acquisition and analysis.	Ensuring the system's generalizability across diverse facial paralysis cases and the need for further validation to confirm the robustness and clinical applicability of the approach.

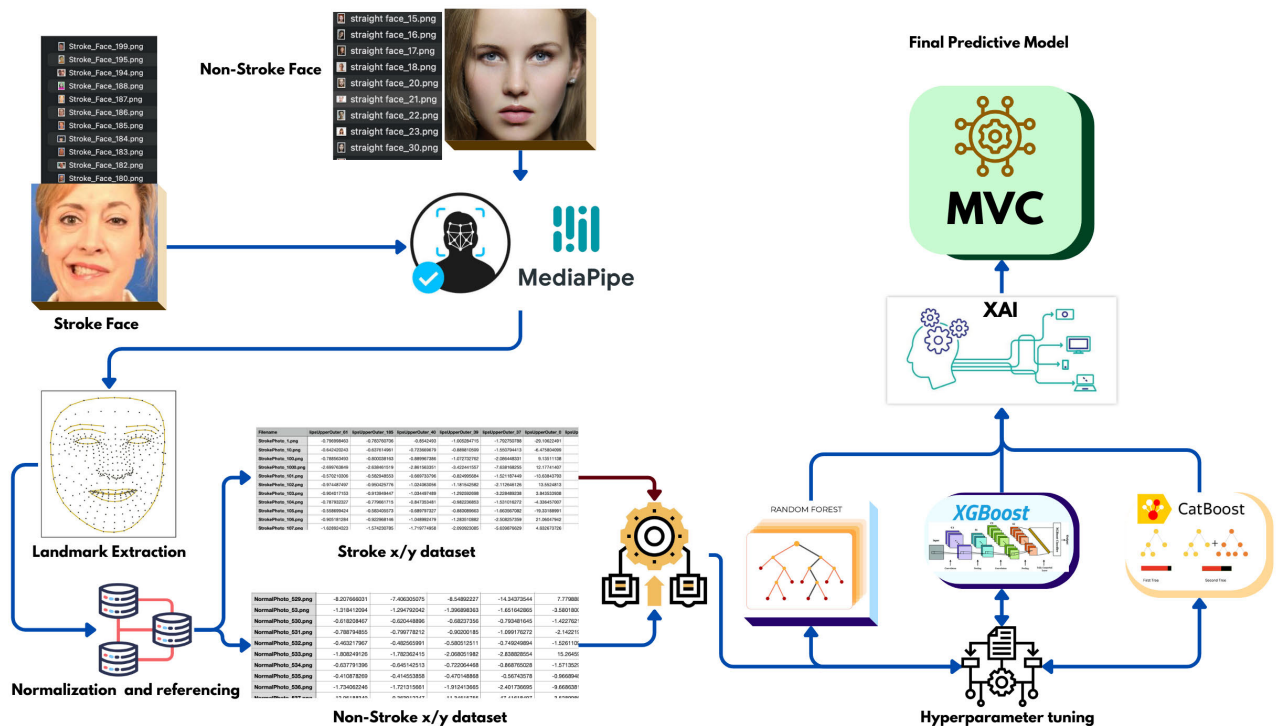
**FIGURE 1.** Overall process flow of facial landmark extraction to model training.

TABLE 2. Demographic data of the collected dataset.

Category	Subcategory	Percentage
Gender	Female	32.80%
	Male	67.20%
Age Group	Teenagers	0.80%
	Middle-Aged	88.40%
	Old Age	17.50%
Ethnicity	White	56.20%
	Latino Hispanic	15.00%
	Asian	13.90%
	Black	10.10%
	Middle Eastern	4.80%

images are normalized using the nose tip as a reference point denoting the origin-like coordinate. The process deducts the nose coordinate from each facial landmark point and scales them accordingly. This ensures that all landmarks are normalized relative to the nose tip as the reference point, thereby reducing variability and enhancing the model's ability to generalize across different facial structures.

The formula for calculating the scale factor, s , between two points, (x_1, y_1) and (x_2, y_2) , can be expressed as the Euclidean distance between them, as shown in (1):

$$s = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}. \quad (1)$$

The formula for normalizing a landmark (x, y) with respect to a reference point (x_q, y_q) , which is the nose tip shown in (2):

$$normalized_landmark = \left(\frac{x - x_q}{s}, \frac{y - y_q}{s} \right). \quad (2)$$

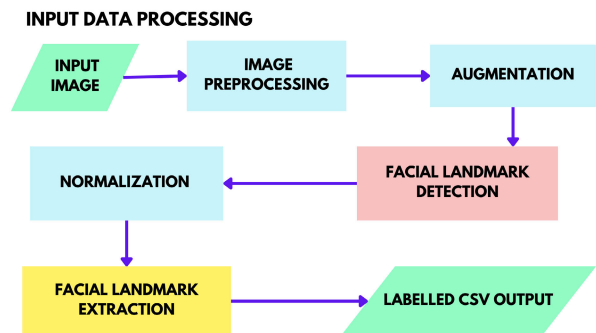
The equation (2) ensures that the reference point becomes the origin (0, 0) after normalization, and the scale factor helps in preserving the relative distances between landmarks. This process standardizes the facial landmarks across different images, ensuring that variations due to image scale or position are minimized. Fig. 6 shows a sample of normalized facial landmarks for both stroke and non-stroke faces.

4) EXPORTING AND LABELING LANDMARKS

Normalized facial landmarks are exported to a structured format as a CSV file for further processing and analysis. Each entry in the CSV file contains the coordinates (x, y) of facial landmarks along with a corresponding label indicating whether the face exhibits stroke symptoms or not. The labeling process is crucial for training and evaluating the machine learning models. We have two classifications: stroke and non-stroke.

B. MODEL TRAINING

The dataset consisted of 2000 facial images, normalized using the nose tip as a reference to standardize the scale and position of features across different faces. 80% of the total images are used for training, and 20% are used for testing randomly. The model training and testing are conducted in three stages: baseline models with default parameters, enhanced models with hyperparameter tuning, and finally, XAI-refined enhanced models with selected important features.

**FIGURE 2.** Diverse stroke patient samples from the dataset.**FIGURE 3.** Process of landmark extraction with mediapipe and exporting dataset.

1) MODEL TRAINING

The system begins with training baseline models: RF, XGB, and CB, along with a MVC that combines their predictions through soft voting, leveraging the strengths of each model. Initially, these baseline models are trained and tested without any hyperparameter tuning, providing a foundation for comparison and further enhancement.

RF: Renowned for its robustness, the RF model is adept at handling noisy data and capturing a wide range of features. By aggregating the results of multiple decision trees, it reduces variance and contributes stable, consistent predictions to the ensemble [27].

XGB: XGB, with its gradient boosting framework, excels at uncovering complex patterns and interactions within the data. Its role in the MVC enhances the ensemble's ability to capture subtle relationships, particularly beneficial in distinguishing nuanced cases.

CB: CB is designed to handle categorical variables effectively and minimize overfitting. Its contribution to the MVC offers a refined perspective, enhancing the ensemble's overall accuracy, especially in distinguishing between closely related categories like "Stroke" and "Non-Stroke".



FIGURE 4. Facial regions and corresponding indexes: Lips (43 indexes), Eyes (114 indexes), Eyebrows (29 indexes), Nose (4 indexes), Cheeks (38 indexes).

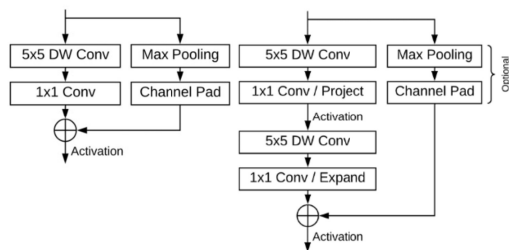


FIGURE 5. BlazeBlock and double BlazeBlock (b) reprinted from Bazarevsky et al. [26].

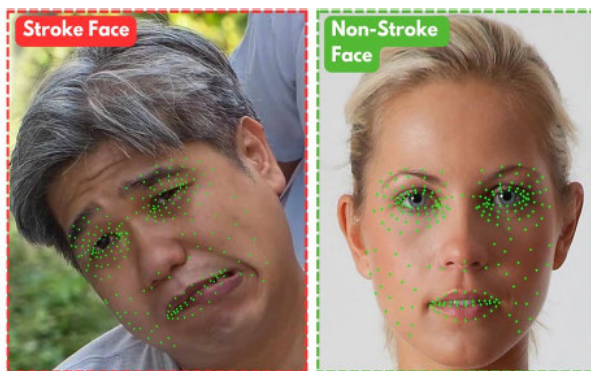


FIGURE 6. Normalized facial landmarks in stroke face and non-stroke face.

Fig. 7 and Fig. 8 show the overview of the training and testing process.

2) ENHANCED MODEL

In this stage of the model training, each model undergoes intensive and rigorous hyperparameter tuning by using Ray Tune. This optimization step significantly enhances the robustness, generalizability and overall accuracy of each of the models. The MVC is also upgraded with these tuned models which further strengthens the system's predictive capabilities.

3) XAI REFINED ENHANCED MODEL

In the last stage, XAI Refined Enhanced Models are introduced which apply XAI methodologies to evaluate the significance of different features. By finding and choosing the most important features each model is retrained using only

the most important data points. This makes the models even more accurate and interpretable. This refinement ensures that the model focuses on the key factors that contribute the most to its predictions. The outcome is the MVC, which not only improves accuracy but also increases transparency in decision-making by emphasizing the role of important features.

In order to develop highly accurate prediction systems for stroke detection, this technique guarantees a development from basic model training to fine-tuned, explainable systems. The comprehensive process of improving the performance of three machine learning models— RF, XGB and CB through hyperparameter tuning with Ray Tune is shown in Fig. 9. In the initial stage, models are trained with baseline configurations and afterwards hyperparameter tuning is employed to optimize their performance and enhance predictive accuracy.

The hyperparameter tuning is carried out using Random Search, Grid Search and Ray Tune iteratively by finding out which search method performs best among the three. Ray Tune is a search method designed to efficiently explore the hyperparameter space and has been found to be the most optimal in this regard. It examines combinations from a predefined search space across 228 features of 29 regions of a facial image to identify the optimal configuration for each model. Each parameter is then manually fine-tuned to improve accuracy. This process allows for the adjustment of parameters like the number of trees, tree depth, learning rates and regularization methods to enhance each model's robustness and prevent overfitting [28].

The application of hyperparameter tuning using Ray Tune significantly increases the predictive capabilities of the individual models and increases their ability to classify complex patterns in the dataset. After tuning the models, they are combined into an enhanced MVC which leverages the complementary strengths of RF, XGB, and CB. This ensemble method takes advantage of the best aspects of each model and makes the method of classification more impartial and accurate.

Improved performance in real-world applications, especially medical diagnosis, is achieved by optimising model configurations and fine-tuning decision boundaries, which greatly increases the predictive ability of both the individual models and the final ensemble.

TRAINING

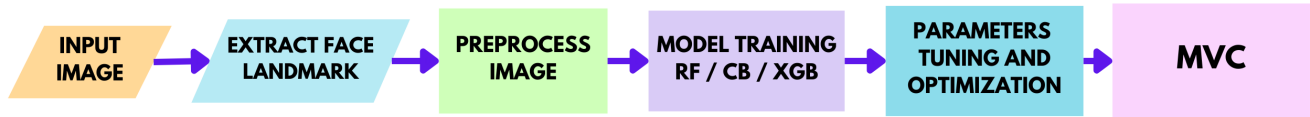


FIGURE 7. Process of training the models RF, CB, XGB and MVC.



FIGURE 8. Overview of the stroke detection flow.

XAI techniques are implemented into the model to improve both interpretability and performance, as given in Fig. 10. XAI plays a critical role by providing transparency into how models make predictions, particularly in understanding the contribution of each facial region to stroke detection. It begins by analyzing 228 facial landmarks and identifying the most relevant ones, focusing on regions prone to asymmetry, such as the eyes, lips, and cheeks. These regions are refined based on their importance across enhanced RF, XGB, and CB models.

XAI allows the system to not only improve prediction accuracy but also to explain which features drive these predictions, thereby making the models more trustworthy and interpretable. By extracting feature importance, XAI enhances the model's ability to focus on the most significant facial regions, leading to better optimization and tuning. In this study, XAI is employed to extract the top 10 significant regions, which reduces the number of important landmarks to 115. Section III-E describes the techniques used to reduce the features. This results in a more targeted and efficient training process, culminating in a MVC that delivers a reliable final prediction based on majority voting from the models.

The model hyperparameter tuning details are discussed in the following section.

4) MODEL OPTIMIZATION

Ray Tune is employed to optimize hyperparameters for training machine learning models across 228 features by efficiently exploring a predefined search space and evaluating performance on the dataset [29]. It uses trial schedulers like HyperBand to prioritize promising trials and terminate ineffective ones early, while leveraging intermediate training results for informed decision-making. By supporting multiple search algorithms and distributing trial execution across a cluster for parallel processing, Ray Tune enables faster and more efficient identification of the optimal hyperparameter configuration.

a: RF

For the RF model, the optimal hyperparameters identified through Ray Tune include setting *n_estimators* to 300 which

indicates an increase in the number of trees and consequently enhances the model's robustness. The *max_depth* is set at 90 enables the trees to grow to a specified depth for better handling of intricate data patterns. The *min_samples_split* is set to 6 and *min_samples_leaf* is set to 3, both contributing to the model's ability to generalize by preventing overfitting on small data splits [30]. The *max_features* is set to 'sqrt' which reduces the feature quantity considered at each split and thereby enhances diversity among trees. As the *bootstrap* parameter is set to False, this indicates that all data points are used without replacement and makes the model more stable. Additionally, the *criterion* for split decision-making is set to 'entropy' for more informative splits.

b: XGB

For the XGB model, the optimal hyperparameters are identified through Ray Tune by setting *max_depth* to 9 which increases model complexity and its ability to capture intricate patterns. The *min_child_weight* is set to 1 which reduces the minimum sum of instance weights in a child and allows for more complex models. The *learning_rate* is set to 0.2, balancing the trade-off between the number of boosting rounds and model accuracy. The *subsample* is set to 0.8 which prevents overfitting by using a subset of training data in each of the boosting round. The *colsample_bytree* is set to 1.0 which ensures that all features are considered at each split while *gamma* is set at 0 which enables splits to occur without any minimum loss reduction requirement. The *n_estimators* is set to 600 which increases model performance and overall stability. The *use_label_encoder* is set to False to ensure compatibility with the latest XGB versions and the *eval_metric* is set to 'rmse' which is suitable for regression tasks. The objective is set to 'binary', specifying the binary classification task.

c: CB

CB is configured and fined tuned in handling categorical data using Ray Tune. It runs 918 iterations through *learning_rate* of 0.293 *depth* of trees has been predetermined at 8 to ensure that complex interaction features may be captured CB and to overfitting regularization, a

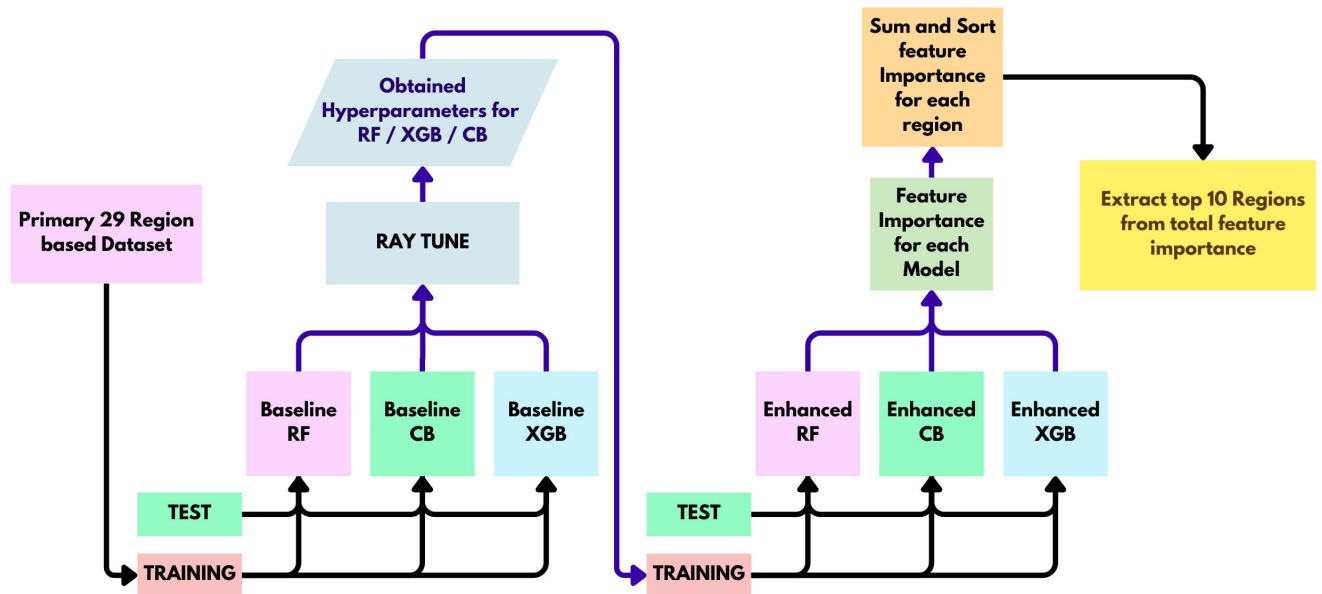


FIGURE 9. Process flow from baseline model to enhanced model and feature extraction.

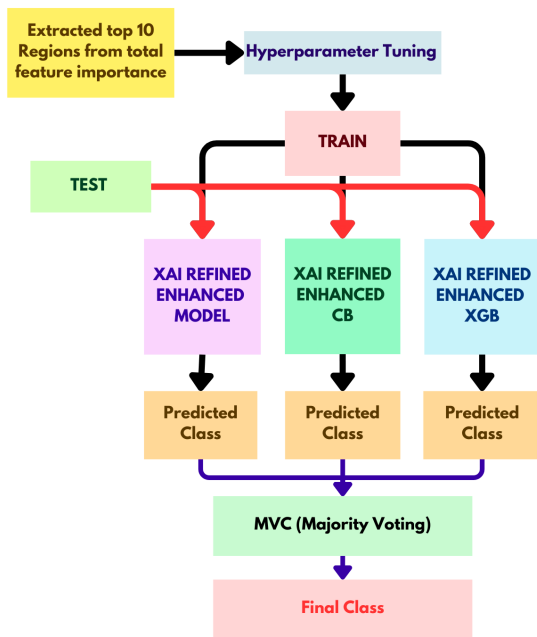


FIGURE 10. Process flow of integrating XAI to refine enhanced models.

bagging_temperature of 0.861 and a *bootstrap_type* of 'MVS' are used. *max_bin* is set to 231 in order to increase the handling of continuous variables, while *scale_pos_weight* is set to 1.925 to stabilize the class weighting in the dataset. Other significant parameters are *colsample_bylevel* at 0.917 for feature selection at each level, *l2_leaf_reg* at 8 for regularization, and *subsample* at 0.648 to further avoid overfitting. Also, *od_type* is set to 'Iter' with *od_wait* of 21 for dynamic overfitting control at training time, while *random_strength* of 0.696 and *min_data_in_leaf* of 9 maintain generalizability.

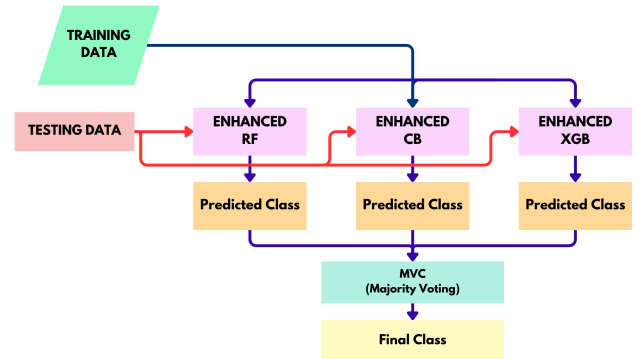


FIGURE 11. Implementation of MVC.

5) IMPLEMENTATION OF MVC

The implementation of the MVC involves combining predictions from three models: RF, XGB, and CB. The MVC uses a hard voting mechanism, where the final prediction is determined by the majority vote of the three classifiers.

Initially, the baseline models are integrated into the classifier where each of the model contributes its own strength to handle complex data patterns. To enhance performance, the models are fine-tuned with the optimal hyperparameters identified through Ray Tune thereby improving the robustness, generalizability, and accuracy of the individual models [31]. At last, XAI techniques are also applied to interpret the results and understand the contribution of each model and feature, helping to further refine the overall classifier's performance. Fig. 11 shows the implementation process of the MVC.

C. EVALUATION OF MODELS

Every algorithm's performance is evaluated using measures like accuracy score, precision, recall, and F1-score [32].

Equations (3), (4), and (5) are used to calculate precision, recall and F1-score respectively. Confusion matrices for each experiment such as the baseline model, enhanced models and XAI refined enhanced model are also analyzed in this study to provide a deeper insight of true positives (TP), true negatives (TN), false positives (FP) along with false negatives (FN).

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (4)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (5)$$

Confusion matrices offer a substantial amount of clarity regarding the number of TP, TN, FP, and FN results and provide insight into the models' ability to differentiate between positive and negative cases. These metrics are crucial for a more comprehensive comprehension of the performance of various models in terms of precision and recall, which directly affect the F1-score.

D. MODEL IMPROVEMENT WITH XAI

Further, in order to enhance both the training process and the interpretability of the MVC with a particular emphasis on the role of facial landmarks, XAI techniques are utilized. XAI provides a profound understanding of how each facial landmark contributes to model performance in detecting stroke-induced facial abnormalities. At first, the extraction of 228 distinct facial landmarks from facial images is done which represents key facial regions such as the eyes, nose, lips, eyebrows, and cheeks—areas that are prone to asymmetry and abnormal movements in stroke patients.

These 228 landmarks are further grouped into 29 specific facial regions. Each landmark is represented by its x and y coordinates, which are normalized to account for variations in facial structure and image scaling. The gradient (y/x) is then computed for each landmark, reducing the two-dimensional coordinates to a single value. This gradient captures proportional changes in vertical and horizontal movements, which are crucial for detecting the facial distortions associated with stroke.

Later, feature importance techniques are applied to three models' RF, XGB, and CB to assess the significance of the 29 facial regions in stroke detection, as shown in Table 3. RF's decision-tree mechanism, XGB's boosting algorithm, and CB's capacity to handle categorical features provide a comprehensive evaluation of feature importance [33].

Algorithm 1 shows the process to calculate feature importance which begins by preparing dataset for analysis through excluding non-relevant columns leaving the attributes necessary for model training. Best hyperparameters for RF, XGB and CB are then initialized, ensuring optimal configuration for each model's architecture. Then each of the models are iteratively assessed to calculate the feature importance. An aggregation structure is also initialized to store cumulative importance scores for each model and facial region. The

features and target variable are partitioned into training and testing subsets. The partitioning is randomized using a seed that varies across iterations (random_state set to 50 times the iteration index), ensuring diverse subsets for robust model evaluation. Feature importance scores are extracted from the models with "feature_importances" attribute. These scores quantify how much each feature contributes to the overall predictive performance of the model. The feature importance scores are aggregated in each facial region by using the predefined facial indexes which are regions consisting facial landmarks. The cumulative score is the sum of the importance scores of all features in the region for each region. The important part of this process is that it is used to capture and quantify the importance of regions as a whole, rather than individual features.

Once the iterative importance computation process is complete for RF, XGB and CB, the facial regions are ranked based on their cumulative importance scores. An iterative subset-wise approach is used for ranking the significant facial regions that analyses the top 29 regions in a facial image and assessing performance metrics for RF, XGB, CB, and MVC. Each iteration consists subsets of regions which are arranged in the following order: {rank_1}, {rank_1, rank_2}, {rank_1, rank_2, rank_3} upto {rank_1, rank_2, rank_3} dots ... rank_29} where "rank_1" is the region with the most aggregated feature importance and "rank_2" is the second most and so on.

The result obtained shows that the highest accuracy is produced by the top 10 regions by the MVC. This ranking enables the identification of the most significant regions that contribute to the target variable. From this ranking, the top 10 regions from a facial image are selected as the most influential regions for subsequent analysis. These top 10 regions accumulate 115 distinct landmarks out of a total of 229 extracted by Mediapipe which underscores their substantial contribution to stroke prediction accuracy. This focused subset not only enhances computational efficiency but also achieves the highest model performance, as validated through key evaluation metrics.

The top 10 significant regions from the aggregated feature importance are: *rightCheek*, *leftCheek*, *lipsUpperOuter*, *lipsUpperInner*, *lipsLowerInner*, *lipsLowerOuter*, *rightEye-browUpper*, *rightEyeLower3*, *leftEyeLower0*, and *leftEye-browUpper*. These regions consist of a total of 115 facial landmarks. These regions are used for further optimization to create a more targeted and efficient feature set.

The highlighted regions and their associated landmarks provide meaningful insights into clinical practice. For example, the important role of certain facial features in the model's decision-making process aligns with observable symptoms of stroke, such as asymmetry in facial tone, eye closure, and smile lines. By mapping these features back to clinical observations, the model's findings can be effectively translated into actionable diagnostic tools, aiding healthcare professionals in early detection and assessment. In Fig. 12,



FIGURE 12. Step by step feature extraction from input image and feature reduction by employing XAI.

the landmark extraction and XAI implementation process for three stroke patients' is illustrated, starting from the input images, followed by landmark extraction using MediaPipe, facemesh construction with 229 landmarks for better facial region visualisation and finally the extraction of the top 10 regions highlighted through XAI.

The next step involves hyperparameter tuning of the RF, XGB, and CB models again using ray tune, specifically focusing on the top 10 significant regions extracted from the subset-wise iterative approach. Adjustments are made to parameters in RF, XGB and CB using ray tune. Then after achieving optimal values, the hyperparameters are again fine tuned empirically to achieve improved metrics [35]. This tuning refines model performance, ensuring greater accuracy and computational efficiency. Finally, the optimized RF, XGB, and CB models are combined into a MVC, utilizing a hard voting mechanism to balance the strengths of each model. This ensemble approach improves prediction reliability, particularly in challenging cases where individual models may falter. The final model not only demonstrates improved performance but also provides key insights into the facial regions most indicative of stroke, enhancing both its interpretability and clinical relevance.

E. REAL-TIME PERFORMANCE ANALYSIS

The real-time performance of the proposed MVC model is crucial in a clinical context where timely and precise decision-making is very important. The MVC is deployed within a Python-based graphical user interface (GUI) application, which lets users to upload an image for stroke diagnosis from facial image as shown in fig. 13. The application preprocesses the image and utilizes the MVC to diagnose stroke. During this process, key performance metrics, including preprocessing time (ms), inference time (ms), and total time taken (ms), are measured to evaluate the model's efficiency.

The robustness and compatibility of the model is tested across a range of hardware systems. Apple M2 Processor, AMD Ryzen 7 6800H, Intel Core i5-7500 and Intel Core i5 4460 is utilized to assess the performance metrics of the model in real-time application.

Benchmarks show the behavior of the model on different use cases with different resources required to execute the model in various computational conditions. The performance indicators allow them to pinpoint the problems in preprocessing or inference in order to see whether the model respects the timing constraints for clinical use. These findings can be used to make assessments regarding the viability of integrating the model within clinical operations given the practical efficiency and reliability demands of such initiatives. The study then shows that the model is practicable and robust by rigorously analyzing real-time performance, assuring it meets the needs of clinical settings, with the highest accuracy and efficiency. Such evaluations are indispensable for closing the gap between research and practice in health care.

IV. EXPERIMENTAL RESULTS

We have trained the models with 80% of the total dataset and tested with 20%. To achieve the best accuracy we have gone through rigorous model testing with different experiments. Experiment 1: Baseline Performance evaluation which tests the models with their default parameters. Experiment 2: Enhanced Performance evaluation which tests the models with Hyperparameter tuning. These two experiments are evaluated with 29 features consisting of 228 features. Experiment 3: Refined Enhanced Performance evaluation which tests the models with reduced features and further tuning.

A. EXPERIMENT 1: BASELINE PERFORMANCE (WITHOUT HYPERPARAMETER TUNING)

RF, XGB, CB with 228 features are tested to classify stroke and non-stroke faces. Random state of 42 and a test data size of 20 percent and training data size of 80% for all three models are applied to achieve the baseline accuracy of the three models. By default, the XGB model uses a learning rate of 0.3 and 100 boosting rounds (trees) as the number of iterations. The optimization method used in XGB is based on the gradient boosting framework with a default objective function of binary logistic regression for classification. RF does not use a specific learning rate as it relies on constructing decision trees, and its default setting involves growing 100 trees. The CB model, on the other hand, has a default learning rate of 0.03 and uses 1000 iterations (or boosting rounds) for optimization. CB also uses a gradient-based optimization algorithm. All three models utilize their respective internal optimizers to minimize errors, and the MVC aggregates the predictions from all these models using a hard voting strategy. The ensemble approach allows for comparison of model performance, particularly accuracy and ROC AUC score.

Algorithm 1 Feature Importance Analysis and Model Training

```

1: Input: Load dataset df
2: Features: features  $\leftarrow df.drop([Filename, is\_stroke\_face], axis = 1)$ 
3: Initialize: Hyperparameters for RF, XGB, and CB models
4: Define: FACE_INDEXES  $\leftarrow$  Predefined facial regions and corresponding indices
5: Initialize all_importances[RF, XGB, CB]  $\leftarrow 0$  for all regions in FACE_INDEXES
6: for each model  $\in \{RF, XGB, CB\}$  do
7:   for i = 1 to 5 do
8:     (X_train, X_test, y_train, y_test)  $\leftarrow train\_test\_split(features, target, random\_state = 50 \times i)$ 
9:     Fit model on X_train, y_train
10:    importances  $\leftarrow model.feature\_importances\_$ 
11:    for each region  $\in FACE\_INDEXES$  do
12:      for each index  $\in region$  do
13:        all_importances[model][region]  $+= importances[index]$ 
14:      end for
15:    end for
16:  end for
17: end for
18: Sort regions by total_importance
19: Select top 10 regions
20: df_filtered  $\leftarrow df.drop(columns\ of\ non-top\ 10\ regions)$ 
21: for each model  $\in \{RF, XGB, CB, Voting\}$  do
22:   Train model on df_filtered
23:   Evaluate model (accuracy, confusion matrix, ROC)
24: end for

```

TABLE 3. Feature importance of RF, XGB, CB, and summation.

Regions	Features	RF	XGB	CB	Total
rightCheek	19	1.029909	0.760791	0.892501	2.683201
leftCheek	19	0.787695	0.599525	0.773957	2.161177
lipsUpperOuter	11	0.395385	0.418346	0.654809	1.468539
lipsUpperInner	11	0.329450	0.227311	0.356087	0.912849
lipsLowerInner	11	0.294919	0.176291	0.323933	0.795142
lipsLowerOuter	11	0.275048	0.229195	0.257556	0.761799
rightEyebrowUpper	8	0.164805	0.164526	0.221544	0.550875
rightEyeLower3	9	0.131113	0.136718	0.100655	0.368486
leftEyeLower0	9	0.094864	0.189709	0.069158	0.353731
leftEyebrowUpper	8	0.121507	0.127090	0.103829	0.352426
rightEyeLower1	9	0.104034	0.143715	0.086056	0.333804
rightEyeUpper0	7	0.070306	0.180430	0.076574	0.327310
rightEyeLower0	9	0.100692	0.160550	0.060579	0.321821
leftEyeLower3	9	0.104344	0.122949	0.092446	0.319739
rightEyeLower2	9	0.110078	0.140853	0.064719	0.315650
rightEyeUpper2	7	0.103445	0.128841	0.068030	0.300317
leftEyeUpper1	7	0.077822	0.185374	0.035525	0.298722
leftEyeLower2	9	0.094555	0.139283	0.059997	0.293835
leftEyeLower1	9	0.090249	0.108907	0.093595	0.292751
leftEyeUpper2	7	0.090189	0.130770	0.057062	0.278021
rightEyebrowLower	6	0.081828	0.115860	0.070840	0.268528
leftEyebrowLower	6	0.079328	0.079579	0.090016	0.248923
rightEyeUpper1	7	0.083078	0.115162	0.049846	0.248086
leftEyeUpper0	7	0.065802	0.130766	0.049928	0.246496
noseBottom	1	0.034437	0.021638	0.105552	0.161627
midwayBetweenEyes	1	0.028327	0.018577	0.101508	0.148413
noseRightCorner	1	0.030914	0.028081	0.052914	0.111909
noseLeftCorner	1	0.025877	0.019161	0.030783	0.075821

1) MODEL ACCURACY

Combining the performance metrics of the baseline models (RF, XGB, and CB), the MVC achieves an impressive

accuracy of 92.75%. This ensemble method demonstrates a balanced and consistent performance with precision, recall, and F1-scores of 0.93 for both Stroke and Non-Stroke classes,

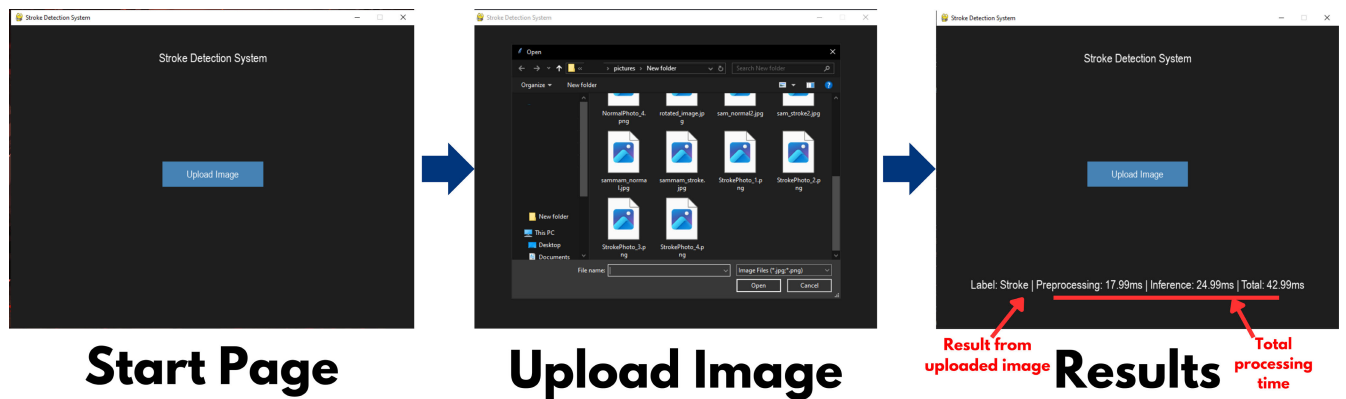


FIGURE 13. GUI of real-time stroke diagnosis with the proposed model.

TABLE 4. Baseline performance summary of RF, XGB, and CB.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
RF	91.5	91.5	91.5	91.5
XGB	92.25	92.5	92	92
CB	92.25	92	92.5	92
MVC	92.75	93	92.5	93

leveraging the strengths of each individual model to deliver reliable and robust classification results.

The individual performance summary reveals that both XGB and CB models each achieve an accuracy of 92.25%, while the RF model shows a slightly lower accuracy of 91.50%. In terms of precision, recall, and F1-scores for “Non-Stroke” class and “Stroke” class, all models exhibit consistent and balanced metrics across both classes. Specifically, the RF model has a recall of 0.90 for non-stroke and 0.93 for stroke class. In comparison, XGB and CB models maintain accuracy and recall values of 0.92 for non-stroke class and 0.93 for stroke class, indicating a slight edge in identifying both non-stroke and stroke cases.

Overall, while RF demonstrates a strong performance, XGB and CB provide a marginally better balance in identifying both stroke and non-stroke cases. The uniformity in F1-scores across all models further underscores their robustness in the binary classification task, with XGB and CB showing a slight advantage in accuracy and class balance. Table 4 shows the summary of classification report for baseline RF, XGB, CB and MVC.

2) RECEIVER OPERATING CHARACTERISTIC (ROC) CURVES

The AUC serves as a crucial metric in this analysis, representing the models’ overall ability to discriminate between the two classes, as shown in Fig. 14. The AUC values for XGB and CB both stand at 0.96, indicating that these models exhibit a high level of accuracy in differentiating between stroke and non-stroke cases. The RF model, with an AUC value of 0.97, shows a marginally superior performance. However, the close proximity of these AUC values suggests that all three models are performing at a very high level, with only subtle differences in their predictive power. Upon closer

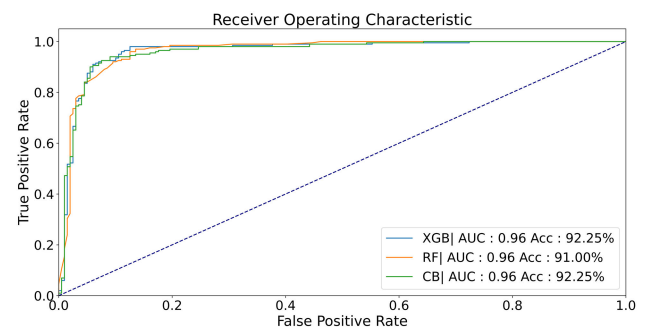


FIGURE 14. ROC curve of baseline performance of XGB, RF, and CB.

examination of the ROC curves, it is observed that all three models produce curves that cluster towards the top-left corner of the plot. This pattern signifies that the models maintain a high TP rate (sensitivity) while simultaneously keeping the FP rate low. This finding implies that each model is effective at correctly identifying stroke cases (high sensitivity) without excessively misclassifying non-stroke cases as strokes (low FP rate). This characteristic is particularly important in medical diagnostics, where balancing the cost of FPs and FNs is critical. The RF model’s curve appears to rise more sharply towards the upper-left corner, especially in the initial detection threshold regions. This suggests that the RF model might be slightly more effective at identifying TPs early on, potentially explaining the slightly higher AUC value observed for this model. Despite this, the differences among the three models are minimal, indicating that XGB and CB also perform robustly across different threshold settings. The comparative analysis reveals that, while the RF model shows a marginal advantage in AUC, the difference is not substantial enough to definitively favor one model over the others based solely on this metric. The close proximity of the ROC curves suggests that all three models—XG, RF, and CB—are highly capable and reliable in their classification tasks.

3) CONFUSION MATRICES

The analysis of confusion matrices generated from the models—RF, XGB, and CB—provides a detailed

understanding of the models' performance in distinguishing between stroke and non-stroke cases.

In the baseline model utilizing the RF algorithm, the confusion matrix shown in Table 5 indicates a balanced performance with an overall accuracy of 91.50%. The TN rate (Non-Stroke correctly identified as Non-Stroke) stands at 45.00%, indicating that nearly half of the Non-Stroke instances are correctly identified. However, the FP rate, where Non-Stroke instances are misclassified as Stroke, is observed at 4.75%. This FP rate suggests a relatively moderate occurrence of misclassification in identifying Non-Stroke cases. On the other hand, the TP rate (Stroke correctly identified as Stroke) is at 46.50%, demonstrating the model's effectiveness in recognizing stroke cases. The FN rate (Stroke misclassified as Non-Stroke) is at 3.75%, indicating a slightly higher tendency to misclassify stroke cases compared to the enhanced model.

For the XGB baseline model, the confusion matrix shown in Table 5 reflects an accuracy of 92.25%, indicating a slightly better performance than the RF baseline. The TN rate is 46.00%, meaning that a slightly higher proportion of Non-Stroke cases are correctly identified. The FP rate is lower at 3.75%, which is an improvement, signifying fewer Non-Stroke cases being misclassified as Stroke. The TP rate, reflecting correctly identified Stroke cases, is 46.25%, which is comparable to the RF baseline but slightly improved. The FN rate is at 4.00%, suggesting a slight increase in misclassification of Stroke cases compared to RF.

The baseline CB model shows an accuracy of 92.25% in Table 5, which is similar to XGB. The TN rate is observed at 45.75%, which is slightly lower than XGB but higher than RF, indicating that the model correctly identifies Non-Stroke cases in nearly half of the instances. The FP rate is maintained at 4.00%, similar to XGB, showcasing a consistent performance in minimizing misclassification of Non-Stroke cases as Stroke. The TP rate, where Stroke cases are correctly identified, is slightly higher at 46.50%, representing the model's strong performance in recognizing stroke cases. The FN rate is at 3.75%, slightly better than XGB, indicating fewer misclassifications of Stroke cases as Non-Stroke.

The confusion matrix for the MVC, which achieved an accuracy of 92.75%, reveals a strong performance in classifying both stroke and non-stroke cases. Specifically, the classifier correctly identified 46.75% of stroke cases (TPs) and 46.00% of non-stroke cases (TNs). However, it also misclassified 3.75% of stroke cases as non-stroke (FN) and 3.50% of non-stroke cases as stroke (FPs), as shown in Table 5. Moreover, the table 5 shows the summary of the confusion matrix for the baseline performance of the three models and the MVC.

B. EXPERIMENT 2: ENHANCED PERFORMANCE (WITH HYPERPARAMETER TUNING)

Further, fine tuning across all three models with hyperparameter tuning and empirical analysis are done and increased

TABLE 5. Confusion matrix summary table for baseline models.

Model	TP (%)	TN (%)	FP (%)	FN (%)
RF	46.50	45.00	4.75	3.75
XGB	46.25	46.00	3.75	4.00
CB	46.50	45.75	4.00	3.75
MVC	46.75	46.00	3.75	3.50

accuracy is achieved. The MVC implementation with all three enhanced models outperformed all the three independent models. Further, fine tuning across all three models with hyperparameter tuning and empirical analysis are done and increased accuracy is achieved. The MVC implementation with all three enhanced models outperformed all the three independent models.

1) MODEL ACCURACY

The application of hyperparameter tuning has significantly refined the predictive capabilities of the models—RF, XGB, CB, and the MVC—leading to noticeable improvements in accuracy, precision, recall, and F1-score. These improvements not only reflect the models' enhanced abilities to classify stroke and non-stroke cases but also reveal deeper insights into the strengths and limitations of each approach.

RF achieved an accuracy of 93.00%, reflecting a robust performance that is slightly more balanced in handling both stroke and non-stroke classifications compared to its baseline performance. The precision of 93% and recall of 94% suggest that the model has become more effective at correctly identifying stroke cases (high recall) while maintaining a strong ability to avoid FPs in non-stroke cases (high precision). This balance indicates that hyperparameter tuning has fine-tuned the decision boundaries of the RF, making it less prone to overfitting and better at generalizing to unseen data.

XGB, with an improved accuracy of 93.50%, demonstrates a marginally superior performance compared to RF. The model's balanced precision and recall of 94% each suggest a consistent ability to minimize both FPs and FNs. This consistency is crucial in medical diagnostics, where the cost of misclassification can be particularly high. The tuning process has evidently optimized XGB's gradient boosting mechanism, enhancing its ability to distinguish subtle differences between the two classes, thus offering reliable predictions in a clinical setting.

CB stands out with an accuracy of 94.00%, marking it as one of the top-performing models in this analysis. The model's precision of 96% for non-stroke cases and a recall of 96% for stroke cases highlight its exceptional ability to correctly identify both conditions. This dual strength in precision and recall suggests that CB is highly effective at drawing nuanced distinctions between stroke and non-stroke patients, minimizing both types of classification errors (FPs and FNs). The superior performance of CB can be attributed to its efficient handling of categorical features and the advanced nature of its boosting algorithm, which appears

TABLE 6. Accuracy summary of enhanced performance of RF, XGB, and CB after hyperparameter tuning.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
RF	93	93	94	93
XGB	93.5	94	94	94
CB	94	96	96	94
MVC	94.25	94.5	94.5	94.5

to have been finely tuned to handle the complexities of medical data in this scenario. Table 6 shows the summarised performance for enhanced performance of the models.

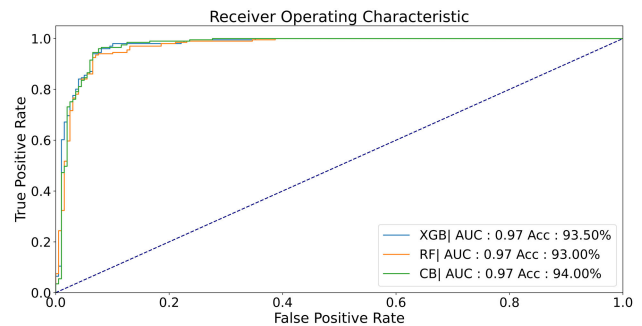
The enhanced and optimized MVC, which integrates the strengths of RF, XGB, and CB, demonstrates a significant improvement in predictive performance, achieving an accuracy of 94.25%. This elevated accuracy suggests that the ensemble method effectively capitalizes on the complementary strengths of the individual models, leading to a more balanced and accurate classification. In terms of precision and recall, the MVC shows a near-equivalent performance for both classes: non-stroke and stroke. For non-stroke cases, the precision is 95%, indicating that out of all the predicted non-stroke cases, 95% are correctly identified. The recall for non-stroke is 93%, meaning that the model successfully detected 93% of actual non-stroke cases. For stroke cases, the precision is 94%, implying that 94% of the predictions labeled as stroke are accurate. The recall is slightly higher at 95%, indicating that the model correctly identified 95% of all actual stroke cases. The F1-score, which balances precision and recall, is consistently high at 94% for both classes, reflecting the model's robustness in handling the classification task.

2) RECEIVER OPERATING CHARACTERISTIC (ROC) CURVES

The ROC curve for the enhanced models after hyperparameter tuning demonstrates a clear and robust improvement across all three models—XGB, RF, and CB, as shown in Fig. 15. The ROC curve is a critical diagnostic tool that evaluates the performance of classification models by plotting the TP rate (Sensitivity) against the FP rate (1-Specificity) across various threshold levels.

For the RF model, it shows substantial improvement, with an AUC of 0.97. This aligns closely with the AUC observed in the XGB model, indicating comparable performance levels. The curve's trajectory indicates that the model achieves a high sensitivity, with a balanced trade-off between TPs and FPs. The model's accuracy is recorded at 93.00%, slightly lower than that of XGB, but still reflecting significant enhancements from the baseline. The RF model's ROC curve suggests that hyperparameter tuning has effectively optimized its performance, especially in minimizing FP rates while maintaining a high level of accuracy.

The ROC curve for the XGB model shows a steep ascent, closely approaching the top left corner of the plot, indicative of high sensitivity with minimal FPs. The AUC for XGB is observed at 0.97, a significant improvement, reflecting the

**FIGURE 15.** ROC Curve for enhanced models with RF, XGB, and CB.

model's enhanced capability to distinguish between Stroke and Non-Stroke cases accurately. The model achieves a balanced and impressive accuracy of 93.50%, confirming its robust performance post-tuning. The nearly perfect linearity of the ROC curve suggests that XGB can maintain a high TP rate while keeping the FP rate to a minimum, reinforcing its reliability in critical decision-making scenarios.

CB, following hyperparameter tuning, exhibits the strongest performance among the models with an AUC of 0.97, matching that of both XGB and RF. The ROC curve for CB shows a nearly vertical ascent towards the upper left corner, signaling excellent discrimination capability between the classes. The accuracy of 94.00% further underscores the model's dominance, achieving the highest accuracy among the three models. This superior performance suggests that CB, with its fine-tuned hyperparameters, offers a slightly better trade-off between sensitivity and specificity, making it a compelling choice for applications requiring high precision and reliability.

3) CONFUSION MATRICES

The analysis of the confusion matrices for the three models—RF, XGB, and CB—after hyperparameter tuning provides valuable insights into their respective performances in predicting stroke and non-stroke cases. Each model's confusion matrix highlights the distribution of TP, TN, FP, and FN, offering a comprehensive view of their classification capabilities.

The RF model, with an accuracy of 93.00%, demonstrates a balanced performance in predicting both stroke and non-stroke cases as shown in Table 7. The confusion matrix shows that 46.00% of the predictions are correctly identified as non-stroke (TN), and 47.00% are correctly classified as stroke (TP). However, 3.75% of the actual non-stroke cases are misclassified as stroke (FP), and 3.25% of actual stroke cases are misclassified as non-stroke (FN). The model's slight tendency to misclassify a small percentage of stroke cases as non-stroke suggests that while it is generally reliable, there remains a marginal risk of underestimating stroke cases, which could be critical in a clinical setting.

The XGB model, achieving a slightly higher accuracy of 93.50%, exhibits strong classification performance similar

TABLE 7. Summary of confusion matrices for enhanced models.

Model	TP (%)	TN (%)	FP (%)	FN (%)
RF	47.00%	46.00%	3.75%	3.25%
XGB	47.00%	46.50%	3.25%	3.25%
CB	48.25%	45.75%	4.00%	2.00%
MVC	47.75%	46.50%	3.25%	2.50%

to the RF model, as described in Table 7. The confusion matrix reveals that 46.50% of the cases are accurately predicted as non-stroke (TN), and 47.00% as stroke (TP). Misclassification rates are slightly lower than those of RF, with 3.25% of non-stroke cases incorrectly predicted as stroke (FP) and an equal 3.25% of stroke cases misclassified as non-stroke (FN). The reduced rate of misclassification for stroke cases indicates that XGB is slightly more precise in distinguishing stroke from non-stroke cases, which is critical for early and accurate intervention.

The CB model, with an accuracy of 94.00%, outperforms the other models by a small margin, indicating its robust ability to classify stroke and non-stroke cases effectively. The confusion matrix indicates that 45.75% of non-stroke cases are correctly identified (TN), and 48.25% of stroke cases are accurately classified (TP) as shown in Table 7. The model's FP rate is 4.00%, slightly higher than XGB, but its FN rate is the lowest at 2.00%. This low FN rate suggests that CB is highly effective in correctly identifying stroke cases, reducing the likelihood of missing potential stroke patients, which is crucial for medical diagnostics. The trade-off is a slightly higher rate of false alarms (FPs), but this may be acceptable given the critical importance of minimizing missed stroke diagnoses.

The MVC, with an accuracy of 94.25%, demonstrates strong performance in distinguishing between stroke and non-stroke cases, as presented in Table 7. The confusion matrix reveals that 46.50% of non-stroke cases are correctly classified (TN), while 47.75% of stroke cases are accurately identified (TP). The model's FP rate is relatively low at 3.25%, indicating minimal false alarms. Its FN rate is slightly higher at 2.50%, meaning a small proportion of stroke cases are missed. However, the model strikes a good balance between minimizing FPs and FNs, which is essential for ensuring both the early detection of stroke cases and reducing unnecessary alerts. The overall high accuracy underscores the effectiveness of this classifier in medical diagnosis scenarios, particularly in stroke prediction.

Table 7 shows the summary of confusion matrices for all the enhanced models.

4) T-SNE PLOT

To visualize the learned feature representations of the classifiers, we applied t-distributed Stochastic Neighbor Embedding (t-SNE), a technique for dimensionality reduction that projects high-dimensional data into a two-dimensional space. This method is particularly effective for visualizing complex patterns and the clustering of data points, providing

insights into how well the model is distinguishing between classes.

The t-SNE plots generated for the RF classifier over 10 simulated epochs (from 50 to 500) in Fig. 16 reveal the model's gradual learning process. Initially, there is significant overlap between the two classes (represented by blue and yellow points), indicating that the decision boundary is not well defined. As training progresses, the clusters begin to show more separation, suggesting that the model is improving in its ability to distinguish between the classes. However, even at later epochs, the boundaries remain vague, with noticeable overlap between the clusters. This overlap implies that the RF model may be struggling with certain boundary cases, which could lead to less reliable predictions.

In contrast, the t-SNE plots for the XGB classifier present a clear and distinct pattern, as shown in Fig. 17. From the very early stages of training, there is a noticeable separation between the two classes (represented by yellow and purple points). This separation becomes more distinct as the model continues to learn. By the time training reaches epoch 200, the classes are well-separated with minimal overlap, indicating that the XGB model is effectively learning a strong decision boundary. The well-defined clusters seen in the t-SNE visualization suggest that XGB is better at capturing the underlying structure of the data, which likely explains its superior performance in classification.

These visualizations highlight a key difference between the models. While the RF classifier shows some ability to distinguish between the two classes, the clusters are less well-defined, indicating potential challenges with classification accuracy. In comparison, the XGB classifier produces much clearer clusters, showing a stronger ability to separate the classes. This aligns with the general expectation that XGB, with its gradient boosting framework, tends to handle complex classification tasks more effectively than RF.

The plots in Fig. 18 show how the decision boundary of the CB evolves over the course of training, highlighting the separation between stroke and non-stroke classes across epochs ranging from 50 to 500. These visualizations provide insight into how the classifier improves its understanding of the data as training progresses. In the early stages, particularly between epochs 50 and 150, a decision boundary starts to form, and a clear separation between the two classes begins to appear. However, some overlap remains, especially near the center of the plot, indicating that the model is still learning to accurately differentiate between stroke and non-stroke cases. As training advances from epoch 200 to 350, the separation between the classes becomes more distinct. The decision boundary is refined, reducing overlap, and showing that the model is getting better at classifying the data, with fewer instances falling near the boundary where errors in classification might occur. By the later stages of training, between epochs 400 and 500, the decision boundary stabilizes, with the classes now being well-separated and minimal overlap remaining. This stabilization suggests that the model has successfully learned the underlying patterns

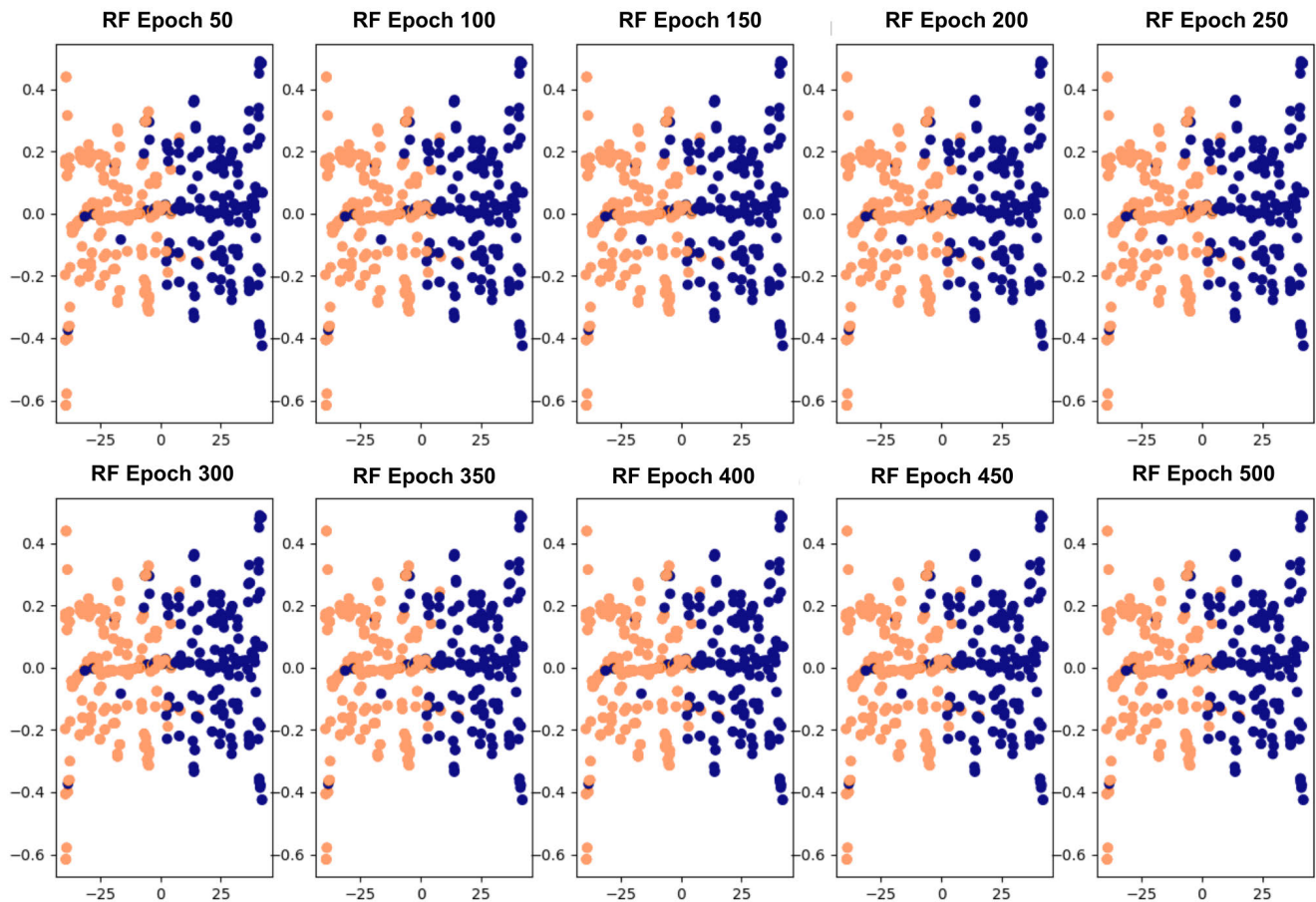


FIGURE 16. t-SNE plot of enhanced RF model over epochs.

in the data, allowing it to make consistent and accurate predictions. The clusters of stroke and non-stroke cases become more distinct, reflecting a higher confidence in the model's predictions.

In the case of the MVC, as depicted in Fig. 19, the early stages of training (from epoch 50 to 150) show the formation of a decision boundary. However, this boundary is still developing, as there is noticeable overlap between the stroke and non-stroke classes, represented by different colors. This overlap indicates that the model is still learning the foundational structure of the data and hasn't yet grasped the complex relationships between features.

As the training progresses, particularly between epochs 200 and 350, the separation between the two classes becomes clearer, and the decision boundary sharpens, reducing the overlap. This suggests that the model is improving its classification ability, learning to more accurately differentiate between stroke and non-stroke cases. The clusters become more coherent, with fewer points near the boundary, which lowers the chances of misclassification.

By the time the model reaches epochs 400 to 500, the separation between the classes becomes stable and more pronounced. The points representing the two classes are

well-separated, and the decision boundary is clearly defined, with minimal overlap. This level of separation indicates that the MVC has successfully learned the complex patterns in the data, resulting in more accurate and reliable predictions. At this stage, the stabilization of the clusters reflects the model's convergence, meaning that further training brings only minor improvements.

These visualizations highlight the varying abilities of the classifiers to learn decision boundaries and distinguish between stroke and non-stroke cases.

C. EXPERIMENT 3: XAI REFINED ENHANCED MODELS

The feature importance scores across 29 regions are analyzed using RF, XGB, and CB. The total importance score is computed, and the top 10 regions identified are as follows:

- rightCheek
- leftCheek
- lipsUpperOuter
- lipsUpperInner
- lipsLowerInner
- lipsLowerOuter
- rightEyebrowUpper
- rightEyeLower3

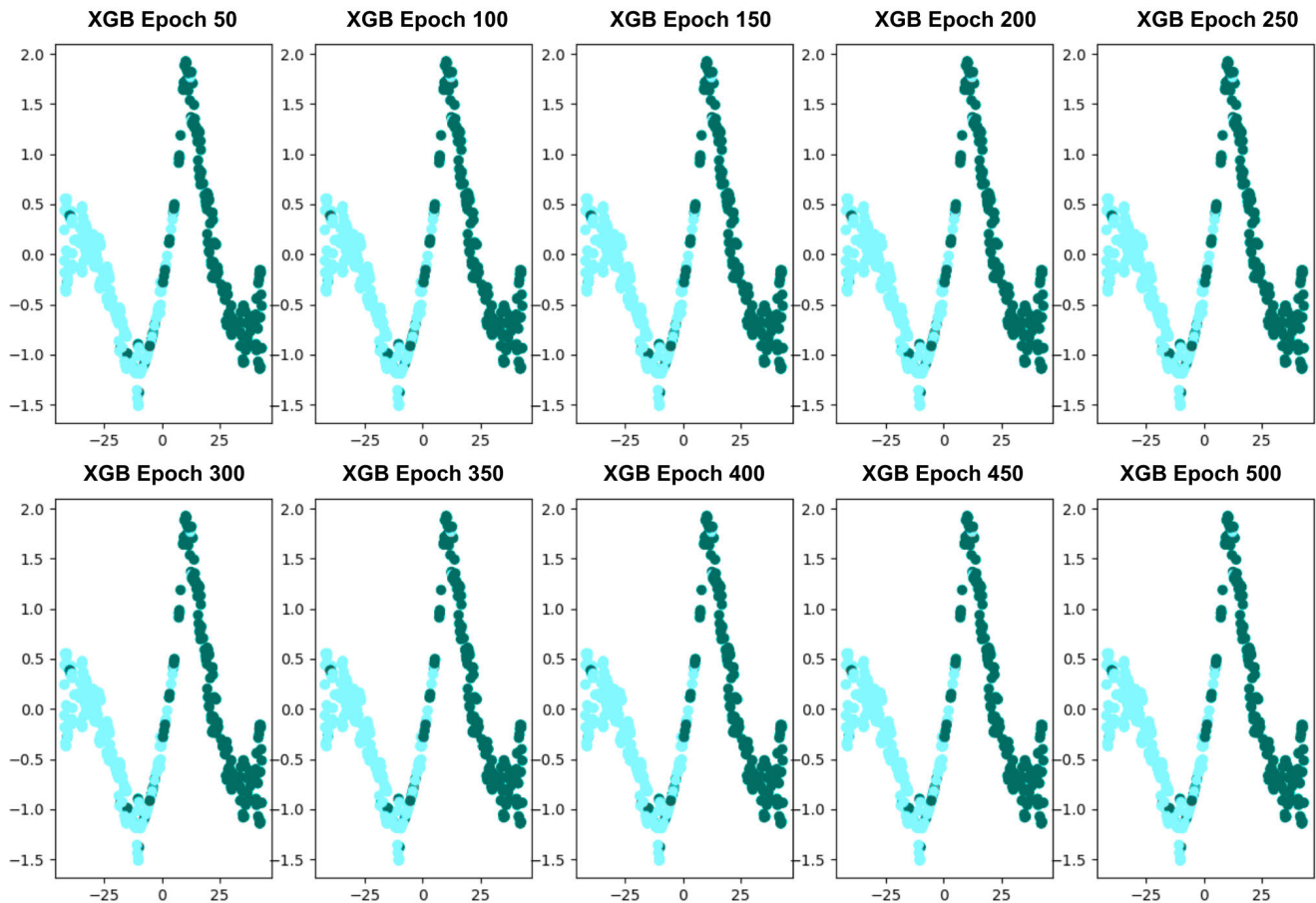


FIGURE 17. t-SNE plot of enhanced XGB model over epochs.

- leftEyeLower0
- leftEyebrowUpper

These regions are found to contribute the most to the models' predictions.

1) MODEL ACCURACY

The RF classifier shows an improvement over XGB, with an accuracy of 93.75%, and 94% for precision, recall, and F1-score as shown in Table 8. The incremental gain in these metrics demonstrates RF's ability to capture more nuanced patterns in the data, likely due to its boosting mechanism, which iteratively enhances model accuracy by focusing on misclassified cases. The improvement in precision and recall suggests that RF is better at both detecting stroke cases and reducing false alarms when compared to XGB.

XGB achieves a solid accuracy of 93.25%, alongside balanced precision, recall, and F1-scores, each standing at 93%. This suggests that XGB is proficient in both correctly identifying stroke and non-stroke cases (precision) and capturing a high number of true stroke cases without too many FNs (recall). The F1-score, reflecting the harmonic mean of precision and recall, shows that XGB maintains a

well-rounded performance. However, while strong, XGB's performance does not quite reach the level of the more advanced models.

CB exhibits the highest individual model accuracy at 94.5%, with a precision of 95%, and recall and F1-score at 94% as shown in Table 8. The increase in precision demonstrates that CB is highly effective at correctly classifying non-stroke cases, reducing the FP rate. Meanwhile, its high recall shows a superior ability to identify true stroke cases, making it especially suitable for critical diagnostic tasks where FNs could have serious consequences. CB's specialized handling of categorical features and its robust performance in real-world settings contribute to these improvements.

MVC which combines the predictive strengths of RF, XGB, and CB, delivers the highest overall performance. With an accuracy of 94.75%, precision, recall, and F1-scores at 95%, the MVC demonstrates the advantage of ensemble methods as shown in Table 8. By aggregating the outputs of multiple models, the MVC minimizes the weaknesses of individual models and maximizes their strengths. This ensures a more balanced and reliable prediction system. The improvement in recall, in particular, reflects the model's ability to more consistently detect true stroke cases, further

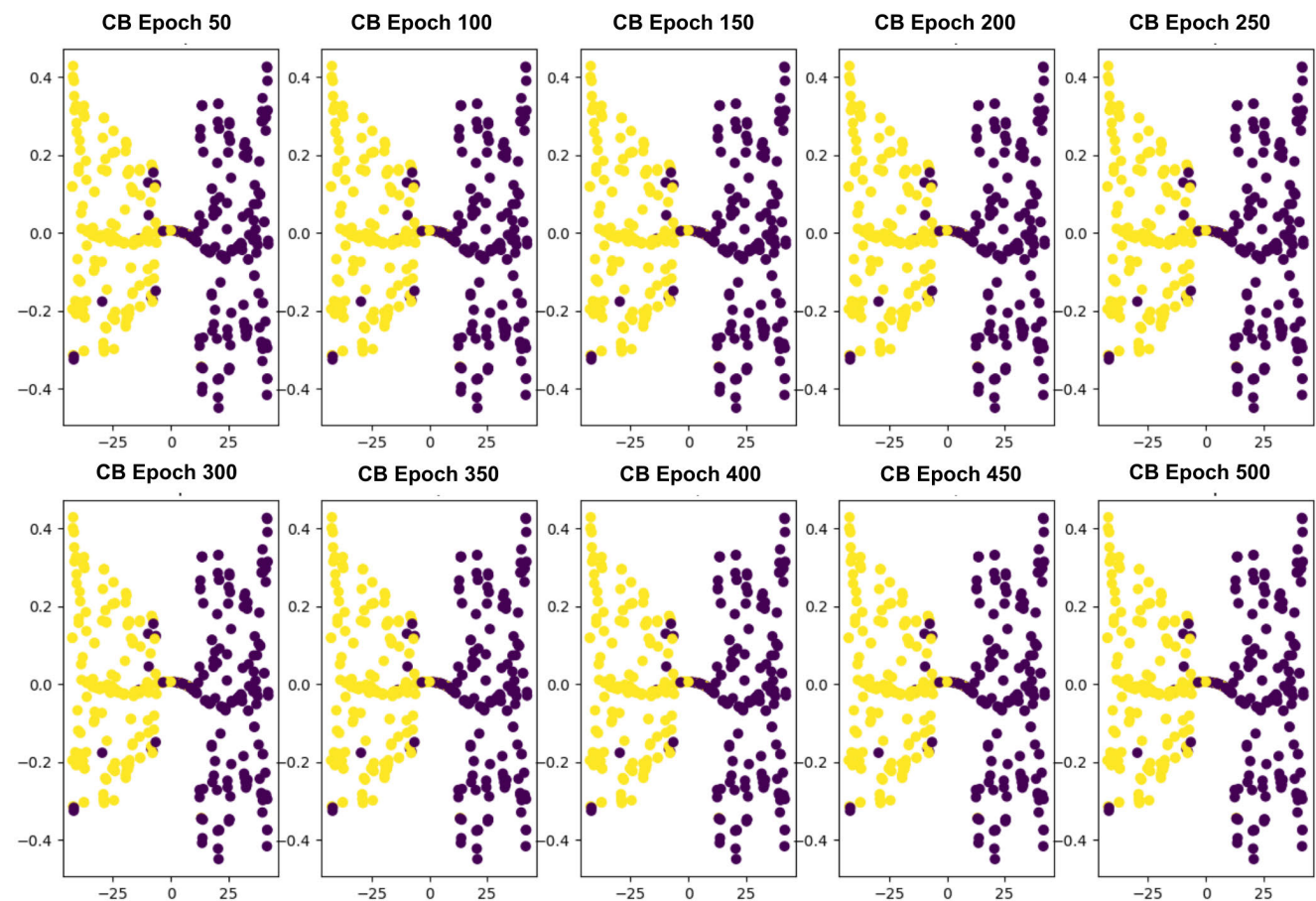


FIGURE 18. t-SNE plot of enhanced CB model over epochs.

TABLE 8. Summary of classification report XAI enhanced refined model.

Model	Accuracy(%)	Precision(%)	Recall(%)	F1-Score(%)
RF	93.75	94	94	94
XGB	93.25	93	93	93
CB	94.5	95	94	94
MVC	94.75	95	95	95

reducing the FN rate. The Multimodal approach, therefore, proves to be the most accurate and efficient for the task of stroke prediction based on facial features, benefiting from the diversity of model outputs to create a more refined overall prediction. The barchart shown in Fig. 20 shows the gradual increase of performance across the three experiments.

Table 9 shows a clear improvement of accuracy for MVC from different stages of model training.

2) RECEIVER OPERATING CHARACTERISTIC (ROC) CURVES

The ROC curve presented in Fig. 21 illustrates the performance of three models—XGB, RF, and CB—after applying XAI techniques and enhanced model refinement. The ROC curve plots the TP Rate (sensitivity) against the FP rate, showing how well each model distinguishes between the stroke and non-stroke classes.

The RF model also displays solid performance with an AUC of 0.96. Although its curve is slightly below the XGB and CB curves, it still shows high discriminatory power. The RF model’s ability to generalize well across the dataset contributes to its 93.75% accuracy. However, its marginally lower AUC compared to XGB and CB suggests that it might generate slightly more FPs or miss TPs in comparison to the other two models.

XGB’s curve demonstrates strong performance, with an AUC of 0.97. The curve closely approaches the upper-left corner, indicating that the model maintains a high TP Rate while minimizing FPs. This performance is reflective of XGB’s boosting technique, which effectively corrects misclassifications by learning from previous errors. The accuracy of 93.25% further supports XGB’s robustness, although it is marginally outperformed by CB in this case.

CB delivers the best overall performance, with the highest accuracy of 94.50% and an AUC of 0.97. Its curve closely matches that of XGB, reflecting a high TP Rate with minimal FPs. The model benefits from its specialized handling of categorical data and its ability to learn from smaller datasets without overfitting. CB’s superior accuracy indicates that it is the most reliable model among the three for this task, consistently making accurate classifications.

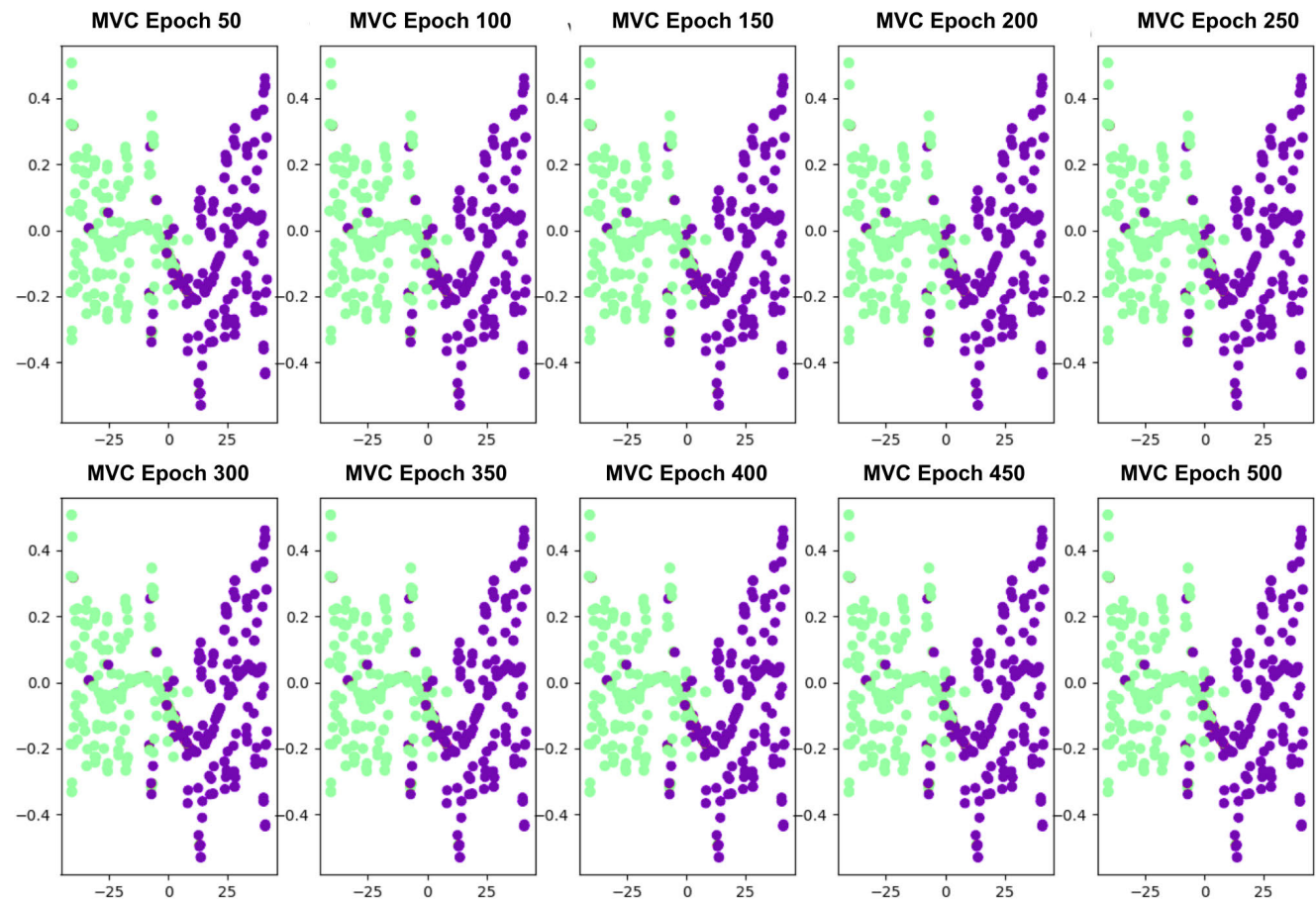


FIGURE 19. t-SNE Plot of MVC over different epochs.

TABLE 9. Comparison table of MVC in different approaches.

Model	Accuracy(%)	Precision(%)	Recall(%)	F1-Score(%)
Baseline MVC	92.75	93.00	93.00	93.00
Hyperparameter Tuned Enhanced MVC	94.25	94.50	94.50	94.50
XAI and Hyperparameter Tuned MVC	94.75	94.50	95.00	95.00

3) CONFUSION MATRICES

The confusion matrices generated from the RF, XGB, CB, and MVC after the application of XAI and the utilization of the top 10 most important facial regions offer significant insights into the models’ performance in diagnosing stroke based on facial features. These matrices provide a valuable lens through which the models’ ability to differentiate between stroke and non-stroke cases can be critically assessed.

The RF classifier demonstrates a strong balance between correctly identifying both stroke and non-stroke cases, as shown in Fig. 22. With a TP rate of 47.75% and a TN rate of 46.00%, it effectively distinguishes between stroke and non-stroke cases. However, the FP rate of 3.75% and the FN rate of 2.50% reveal a minor degree of misclassification.

While this model has a relatively small percentage of stroke cases incorrectly classified as non-stroke, it also has a slightly higher rate of non-stroke cases falsely classified as stroke. This balance, though not perfect, showcases RF’s capability in handling the complex patterns related to stroke prediction from facial features. The classifier’s ability to minimize FNs is crucial in medical diagnoses, where incorrectly diagnosing a stroke case as non-stroke could lead to severe consequences.

The XGB model shows similar performance to RF in Fig. 23, with a TP rate of 47.00% and a TN rate of 46.25}. While it maintains a high level of accuracy, the FP rate of 3.50% and the FN rate of 3.25% suggest a slight compromise in its stability compared to RF. The increase in FNs, though minimal, indicates that a small proportion of stroke cases are misdiagnosed as non-stroke. This might be

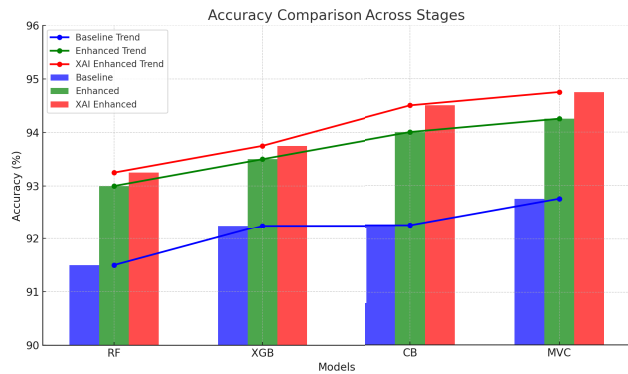


FIGURE 20. Comparison of three experimental stages of RF, XGB, CB and MVC.

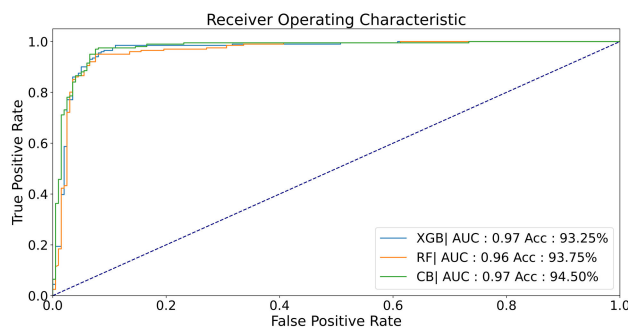


FIGURE 21. ROC curve for XAI refined enhanced models.

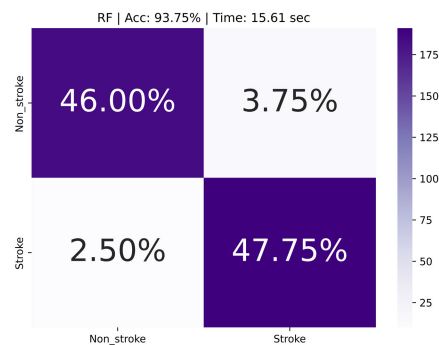


FIGURE 22. XAI enhanced Refined RF confusion matrix.

attributed to the model's sensitivity to small variations in the facial regions. Nevertheless, XGB continues to perform well overall, particularly in its ability to manage non-stroke cases. Its performance remains robust but slightly less consistent compared to RF in critical stroke detection scenarios.

CB emerges as an effective model with a remarkable TP rate of 48.75% and a TN rate of 45.75%. It also has the lowest FN rate of 1.50% among the individual models, as shown in Fig. 24. This is particularly noteworthy in medical diagnostics, as minimizing FNs is vital in ensuring that stroke cases are not overlooked. Despite a marginally higher FP rate of 4.00%, which indicates that a small portion of non-stroke cases are mistakenly classified as stroke, the model excels in detecting stroke cases. CB's ability to detect subtle variations

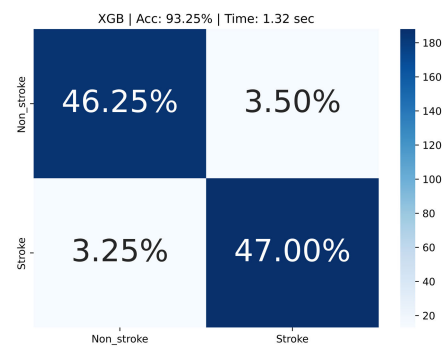


FIGURE 23. XAI enhanced refined XGB confusion matrix.

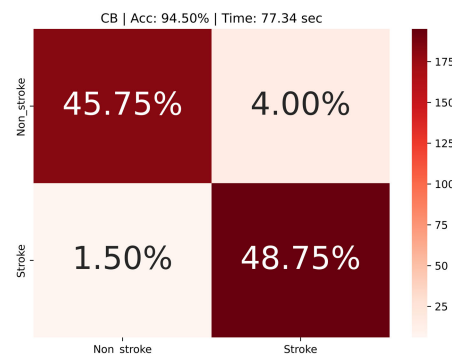


FIGURE 24. XAI enhanced refined CB confusion matrix.

in facial features related to stroke may explain its superior performance in this area. By correctly identifying a higher percentage of stroke cases while maintaining a strong balance with non-stroke identification, CB demonstrates its advanced capabilities in handling complex diagnostic tasks.

The MVC, which aggregates the strengths of RF, XGB, and CB, provides the best overall performance, as shown in Fig 25. With a TP rate of 48.50% and a TN rate of 46.25%, it successfully balances the correct identification of stroke and non-stroke cases. Furthermore, the FP rate of 3.50% and the FN rate of 1.75% demonstrate the model's ability to reduce misclassifications on both sides. By combining the predictive power of multiple models, the MVC reduces the errors that might arise in individual models. This makes it the most reliable classifier for stroke detection, achieving the highest accuracy with fewer FPs and negatives. The integration of XAI and the MVC's hybrid nature allow it to effectively handle the complexity of stroke diagnosis from facial landmarks. The Table 11 shows the comparison between different methods applied for stroke diagnosis.

D. REAL-TIME PERFORMANCE METRICS ON DIFFERENT SYSTEMS

The performance of the proposed MVC model in real-time is critical in a clinical setting where timely and accurate decision making is critical. The model is deployed in a Python based GUI application and is used to evaluate its efficiency by using facial images uploaded by the users to diagnose

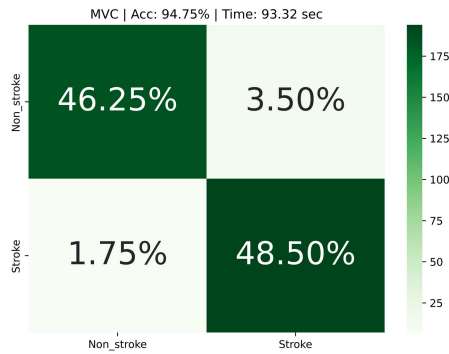


FIGURE 25. XAI enhanced refined MVC confusion matrix.

TABLE 10. Real-time performance benchmarks of the model.

Processor	RAM (GB)	Preprocessing Time (ms)	Inference Time (ms)	Total Time Taken (ms)
Apple M2 Chip	16	31.173	8.21	39.398
AMD Ryzen 7 6800H	16	23.0	30.0	53.0
Intel Core i5-7500	16	30.0	40.0	70.0
Intel Core i5-4460	8	40.0	40.0	80.0

stroke. Real-time performance benchmarks are conducted on different hardware configurations such as Apple M2, AMD Ryzen 7 6800H, Intel Core i5-7500 and Intel Core i5-4460 to validate the model’s robustness and compatibility as shown in table 10. These tests demonstrate the ability of the model to deal with different computational conditions relevant for clinical applications. Moreover, Memory speed, disk access time and CPU efficiency play the largest role in determining the amount of processing time taken to pre process the image, including loading, normalization and feature extraction.

The proposed MVC is applied in a range of hardware systems from high configuration to resource constrained ones. Table 10 shows different performance metrics of the MVC for different processor types and memory capacity. Due to high speed architecture and efficient memory utilization, the Apple M2 chip processor of Macbook Pro M2 appears to exhibit the lowest total processing time of 39.398 ms. However, the Intel Core i5 4460, with lesser RAM, takes 80 ms in total, demonstrating on the one hand that hardware constraints can have an impact on processing time of both pre-processing calls and inference calls. However, the model can be used for real-time stroke diagnosis on all tested configurations, indicating its robustness and adaptability in use in various computational conditions.

E. COMPARISON WITH OTHER APPROACHES

The proposed model demonstrates superior performance compared to others due to several key advancements in methodology and dataset usage. Gebre et al. [8] employ

a user-controlled model based on color matching and supervised learning, which heavily relies on manual input and is less effective under non-ideal video conditions, leading to potential inaccuracies in stroke detection as shown in Table 11 Aldridge et al. [9] apply machine learning techniques using HoG features and penalized Linear Discriminant Analysis, but their model is trained on a relatively small dataset of only 189 videos, limiting its ability to generalize across diverse cases of facial asymmetry. Barbosa et al. [35] utilize a hybrid classifier for facial paralysis assessment, but the dataset consists of only 65 subjects, making the model less robust and potentially prone to overfitting. In comparison, the proposed model utilizes a larger, augmented dataset of 2000 facial images, ensuring better generalization across a wider range of stroke and non-stroke cases. MVC combines the strengths of RF, XGB, and CB, addresses the limitations of single-model approaches seen in other methods. Furthermore, XAI techniques increases model transparency by finding major facial landmarks which is most relevant to stroke detection also improves accuracy and interpretability. Thorough hyperparameter tuning further improves the performance of the proposed model which leads to a significantly higher accuracy (94.75%) compared to the models by Gebre et al. (88.06%), Aldridge et al. (88.90%), and Barbosa et al. (94%). These factors collectively contribute to the proposed model’s improved reliability and effectiveness in stroke detection.

V. DISCUSSION

The method proposed in this study diagnoses stroke patients by analysing facial landmarks obtained using MediaPipe with machine learning models like RF, XGB, and CB. The proposed MVC with XAI techniques applied gives an interpretable, lightweight and accurate solution for stroke detection. XAI further improves transparency of the model and helps refining it by deciding which facial regions are the most significant. Such developments bring out to the fore the potential of the proposed system as a cheap, non-invasive diagnostic solution.

Testing is also done across different devices with high performance systems, as well as basic machines which show the system possessing a flexible nature. Since GPU acceleration is not used for real-time inference testing, the MVC is suitable for resource-constrained deployments. However, GPU-based technology would speed up inference and minimise processing time, improving model efficiency. The model can be used in clinical environments for early stroke detection in situations with fewer resource. This rapid and reliable output of the device improves clinical care by enabling prompt treatment decisions which may reduce stroke associated mortality rates. Future work might involve real-time inference with the MVC implemented using cloud technology, which can use GPU-accelerated systems and be used in remote and resource constrained places with high precision and low latency.

TABLE 11. Comparison of state-of-the-art stroke detection methods.

Reference	Method and Contribution	Dataset	Precision (%)	Recall (%)	F1-Score (%)	Accuracy (%)
Gebre et al. [8]	Color matching algorithms and supervised learning for gesture stroke detection under non-ideal conditions	Annotated video data of a young woman gesturing, 3.6 minutes long, 60 strokes	-	-	-	88.06
Aldridge et al. [9]	HoG features with penalized Linear Discriminant Analysis for detecting unilateral facial weakness	Videos of people with unilateral facial weakness (n=93) and normal smiles (n=96), 189 videos total	-	-	-	88.90
Barbosa et al. [35]	Hybrid classifier using iris segmentation and key-point detection for facial paralysis classification	Facial images of 65 subjects (50 patients, 15 healthy subjects), 325 facial images total	-	-	-	94
Veeravalli & Bodapati [11]	CNN with key-point generation (MediaPipe) and Partial Curve Mapping for detecting facial asymmetry	Facial images of individuals with facial palsy, showing droopy faces and asymmetry	-	-	-	90
Proposed Model	MediaPipe extracted facial landmark analysed MVC with XAI	1000 facial images (500 stroke, 500 non-stroke) from public datasets, augmented to 2000	94.50	95	95	94.75

However, this has some limitations, despite of the promising results. In dim surroundings the system operates with lower precision as this affects detection of face landmarks. Secondly, the model requires human validation for stroke detection because it is a sensitive task and there is the risk of misdiagnosis. It may also struggle with subtle or ambiguous facial asymmetries more over, and the system is limited to pre-processed datasets and facial expression change detection limits its practical use in various population segments.

To optimize the system, several approaches can be thought of. Transformer based architectures may enhance the model's capacity to process sequentially data and raise the model's classifying capability. Moreover, the use of large language models (LLMs) to enhance precision and scalability on analyzing multimodal data can be also implemented. By implementing fine-tuned Large Vision Models (LVMs) with the proposed model, A highly accurate and context-sensitive diagnostic framework that utilizes advanced visual feature extraction to strengthen robustness, enhance versatility across various datasets, and optimize performance in complex real-world clinical settings can be developed. It's also significant to expand the dataset to include a wider variety of facial images and tackle low-light performance challenges to bolster the model's robustness. Furthermore, the automated calibration techniques could be incorporated and the system real time capability could be enhanced to further broaden its clinical application.

VI. CONCLUSION

In conclusion, a comprehensive methodology for early stroke detection is presented in this research by employing facial

landmark analysis in conjunction with machine learning and Explainable AI (XAI) techniques. The approach primarily uses MediaPipe to extract 228 facial landmarks from facial images that are then normalized and aggregated into key facial regions known to exhibit stroke-related asymmetry, such as the eyes, cheeks, and lips. The curated and augmented dataset, comprising both stroke and non-stroke images is used to train individual models—RF, XGB and CB—whose predictions are combined in an MVC. Rigorous hyperparameter tuning using Ray Tune is conducted to enhance the generalizability and robustness of each of the models, leading to a best-in-class ensemble performance.

It is observed that the MVC is able to attain an accuracy of 94.75%, which exceeds the precision, recall, F1-score of the individual classifiers- RF, XGB and CB even after applying XAI. The regions of the facial region most influential for stroke detection through XAI are around the eyes, cheeks and lips. This finding is in agreement with clinically diagnosed facial asymmetry. T-SNE visualization further details the model's discriminative power to classify stroke affected and non stroke samples and validates the decision boundaries of the model. The proposed system maintains efficient computational performance in real-time performance test and is therefore suitable for deployment in clinical or resource constrained environments. Over and above this, XAI methodologies are also included in the incorporation to make the decision process transparent with the help of critical facial landmarks by revealing the critical facial landmarks to which the classification decision is driven, which is of particular importance in healthcare applications as interpretability is one of the

major requirements. Future work can explore the expansion of the training dataset and optimization of the system to handle the challenging imaging conditions and inclusion of LVMs and LLMs. The proposed framework has large potential in being able to perform early intervention and better management of stroke diagnosis by using a non-invasive, cost-effective method to provide rapid and transparent results.

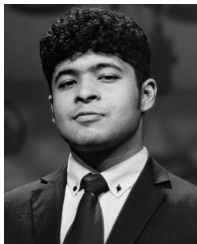
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